The Riemann Problem in Special-Relativistic Magnetohydrodynamics

Lecture notes for a graduate course

with derivations, exercises, and notes on an exact solver in practice

These notes develop the theory of the one-dimensional Riemann problem of ideal special-relativistic magnetohydrodynamics (SRMHD) from the conservation laws up to the exact and approximate solvers used in multidimensional codes. Derivations are given step by step. The algebraic identities marked as such were checked symbolically or numerically with the script `sympy_checks.py` that accompanies these notes (Chapter D). Statements about the reference implementation (the exact solver in `rmhd_final` and the finite-volume code in `HLLD`) are restricted to what its source and its documentation record.

Version of 15 October 2026

Contents

Conventions and notation iv

1 Introduction 1
1.1 Hyperbolic conservation laws and the Riemann problem . . . 1
1.2 Godunov's method . . . 2
1.3 Why relativistic MHD . . . 3
1.4 The exact solver in practice, and a roadmap . . . 3
Exercises . . . 3

2 The equations of ideal SRMHD 5
2.1 Fluid . . . 5
2.2 Electromagnetic field and the ideal-MHD condition . . . 5
2.3 Total energy–momentum tensor and the conservation laws . . . 6
2.4 Conservative form . . . 6
2.5 Equation of state and sound speed . . . 7
2.6 Primitive recovery . . . 8
2.7 Newtonian limit . . . 8
Exercises . . . 9

3 Characteristics and hyperbolicity 10
3.1 The one-dimensional system . . . 10
3.2 Characteristic speeds from a covariant plane-wave analysis . . . 10
3.2.1 The dispersion relation in the fluid rest frame . . . 11
3.2.2 Laboratory-frame speeds . . . 11
3.3 Rest-frame speeds and their ordering . . . 12
3.4 Properties of the laboratory-frame speeds . . . 13
3.5 Degeneracies . . . 14
Exercises . . . 15

4 Discontinuities 16
4.1 Rankine–Hugoniot conditions . . . 16
4.2 Mass flux and the classification of discontinuities . . . 16
4.3 Contact and tangential discontinuities . . . 17
4.4 Three invariants of discontinuities with $J \neq 0$ . . . 17
4.5 Rotational (Alfvén) discontinuities . . . 18
4.6 Shocks . . . 19
4.6.1 The relativistic hydrodynamic Taub adiabat . . . 19
4.6.2 Magnetosonic shocks . . . 19
4.7 Admissibility . . . 20
Exercises . . . 22

5 Rarefaction waves **23**
5.1 Self-similar simple waves . . . 23
5.2 Relativistic hydrodynamics . . . 23
5.3 Relativistic MHD . . . 24
5.4 The state on the ray $\xi = 0$ . . . 25
Exercises . . . 25

6 The relativistic hydrodynamic Riemann problem **26**
6.1 Structure of the solution . . . 26
6.2 The shock branch . . . 26
6.3 The rarefaction branch . . . 27
6.4 The equation for the central pressure . . . 27
6.5 Wave patterns and the relative velocity . . . 27
6.6 Tangential velocities . . . 28
Exercises . . . 28

7 The SRMHD Riemann problem **29**
7.1 Seven waves, eight states . . . 29
7.2 Special cases . . . 29
7.2.1 Purely tangential field orthogonal to the flow . . . 29
7.2.2 Normal field zero: three waves . . . 30
7.2.3 Coplanar data: five waves . . . 30
7.3 The exact solution of Giacomazzo and Rezzolla . . . 30
7.4 An exact solver used as a flux function . . . 31
Exercises . . . 34

8 Non-regular solutions and non-uniqueness **35**
8.1 Non-convexity and compound waves . . . 35
8.2 Coplanar non-uniqueness . . . 35
8.3 Selection by dissipation . . . 36
8.4 What this means for an exact solver . . . 36
8.5 The failure classes of the reference implementation . . . 37
Exercises . . . 37

9 Approximate Riemann solvers **38**
9.1 HLL . . . 38
9.2 HLLC for relativistic MHD . . . 39
9.3 HLLD for relativistic MHD . . . 40
9.4 Roe-type and linearised solvers . . . 41
9.5 Trade-offs . . . 42
Exercises . . . 42

10 Riemann solvers in multidimensional codes **43**
10.1 The finite-volume update . . . 43
10.2 Reconstruction . . . 43
10.3 Time integration . . . 44
10.4 Constrained transport . . . 44
10.5 Standard tests . . . 45
Exercises . . . 46

A Algebra of b^μ and the components of $T^{\mu\nu}$ **47**

B Eigenstructure **48**
B.1 Characteristic matrix in covariant variables 48
B.2 Right eigenvectors 48

C Notation **50**

D What was verified, and how **51**

Conventions and notation

Units and metric. We use units with $c = 1$ and the Minkowski metric $g_{\mu\nu} = \mathrm{diag}(-1, 1, 1, 1)$ in Cartesian coordinates (t, x, y, z), so covariant derivatives reduce to partial derivatives. Greek indices run over $0, \ldots, 3$, Latin indices over $1, 2, 3$; repeated indices are summed. The magnetic field is measured in Heaviside–Lorentz-type units in which the factor $\sqrt{4\pi}$ has been absorbed into $\boldsymbol{B}$: the magnetic pressure is $b^2/2$ and the Maxwell stress has no 4π. (A Gaussian field is $\sqrt{4\pi}$ times ours.) This is the convention of the codes discussed here.

Fluid variables. ρ is the rest-mass density, ϵ the specific internal energy, p the gas pressure, $h = 1 + \epsilon + p/\rho$ the specific enthalpy of the gas, $\boldsymbol{v}$ the three-velocity, $W = (1 - v^2)^{-1/2}$ the Lorentz factor and $u^\mu = W(1, \boldsymbol{v})$ the four-velocity. The laboratory magnetic field is $\boldsymbol{B}$, the comoving field four-vector is b^μ, and

$$\eta \equiv \boldsymbol{v} \cdot \boldsymbol{B}, \qquad b^2 \equiv b^\mu b_\mu, \qquad w \equiv \rho h + b^2, \qquad P \equiv p + \tfrac{1}{2} b^2, \qquad \sigma \equiv \frac{b^2}{\rho h}.$$

w is the *total* enthalpy density, P the *total* pressure and σ the magnetisation. Two warnings about other people's notation: the exact solver of Giacomazzo and Rezzolla (2006) writes h for the *total* specific enthalpy $h_g + b^2/\rho$ and h_g for ours, and Antón et al. (2010) write $\mathcal{E}$ for our w. Where we quote their equations we translate them.

Speeds. c_s is the sound speed, $c_a^2 = b^2/w$ the square of the Alfvén speed in the fluid frame, and $\zeta^2 = c_s^2 + c_a^2 - c_s^2 c_a^2$. Characteristic speeds in the laboratory are written λ, with subscripts e (entropy), A (Alfvén), s (slow), f (fast). Shock speeds are V_s, with Lorentz factor W_s.

Jumps. Across a discontinuity, "a" labels the state *ahead* of the front (into which it moves) and "b" the state *behind*. Following Giacomazzo and Rezzolla (2006) we write

$$[\![Q]\!] \equiv Q_a - Q_b, \qquad (\!(Q)\!) \equiv Q_a + Q_b.$$

Planar symmetry. In the Riemann problem all quantities depend on t and on the coordinate x normal to the initial discontinuity. "Normal" quantities carry the index x (or n), "tangential" ones the indices y, z (or t, as in B_t).

Code references. File names refer to the two repositories of the reference implementation: `HLLD/` (finite-volume code; e.g. `src/physics/hlld.py`) and `rmhd_final/` (exact solver; e.g. `rmhd/batched/planar5_b.py`). Its documentation is cited by file name: `HLLD/docs/GUIDE.md` ("the guide") and `HLLD/docs/what_we_solve.md` ("the solver record").

Chapter 1

Introduction

1.1 Hyperbolic conservation laws and the Riemann problem

Ideal relativistic magnetohydrodynamics, like ideal gas dynamics, is a system of conservation laws. In one space dimension it has the form

$$\partial_t \boldsymbol{U} + \partial_x \boldsymbol{F}(\boldsymbol{U}) = 0, \tag{1.1}$$

where $\boldsymbol{U} \in \mathbb{R}^m$ collects densities of conserved quantities and $\boldsymbol{F}(\boldsymbol{U})$ their fluxes. The system is *hyperbolic* if the Jacobian $\mathsf{A}(\boldsymbol{U}) = \partial \boldsymbol{F}/\partial \boldsymbol{U}$ has real eigenvalues and a complete set of eigenvectors for every admissible $\boldsymbol{U}$; the eigenvalues are the speeds at which small disturbances propagate. Solutions of (1.1) develop discontinuities in finite time even from smooth data, so they are understood in the weak (integral) sense,

$$\int_{x_1}^{x_2} \boldsymbol{U}(x,t_2)\,\mathrm{d}x = \int_{x_1}^{x_2} \boldsymbol{U}(x,t_1)\,\mathrm{d}x - \int_{t_1}^{t_2} \left[\boldsymbol{F}(\boldsymbol{U}(x_2,t)) - \boldsymbol{F}(\boldsymbol{U}(x_1,t))\right]\mathrm{d}t, \tag{1.2}$$

supplemented by admissibility conditions that select the physically relevant weak solution (Chapter 4).

Definition 1.1 (Riemann problem)**.** The Riemann problem for (1.1) is the initial-value problem with piecewise constant data

$$\boldsymbol{U}(x,0) = \begin{cases} \boldsymbol{U}_L, & x < 0, \\ \boldsymbol{U}_R, & x > 0. \end{cases} \tag{1.3}$$

Both (1.1) and (1.3) are invariant under the scaling $x \to \alpha x$, $t \to \alpha t$ with $\alpha > 0$. If the (admissible) solution is unique, it must therefore be invariant too, $\boldsymbol{U}(x,t) = \boldsymbol{U}(\alpha x, \alpha t)$, i.e. it depends on x and t only through the similarity variable

$$\xi \equiv x/t. \tag{1.4}$$

The solution is a fan of *waves* emanating from the origin and separating constant states. A wave is either a discontinuity moving at constant speed (shock, contact, rotational discontinuity) or a centred *rarefaction*, a smooth self-similar fan. For an m-component strictly hyperbolic system with genuinely nonlinear or linearly degenerate fields, the classical theory (Lax, 1957) gives, for data close enough together, exactly one wave per characteristic family and $m-1$ intermediate constant states. Ideal MHD violates the hypotheses of that theorem at some states, which is the source of most of the difficulty treated in these notes.

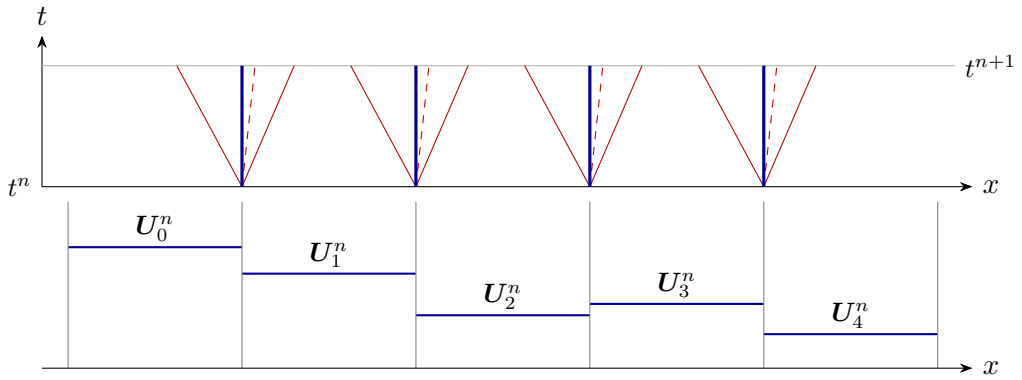


Figure 1.1: Godunov's method. Bottom: cell averages define piecewise-constant data at t^n. Top: each interface carries a Riemann problem; the solution on the interface ($\xi = 0$, thick line) is constant in time and gives the flux (1.7).

1.2 Godunov's method

Divide the line into cells $C_i = [x_{i-1/2}, x_{i+1/2}]$ of width Δx and store cell averages $\boldsymbol{U}_i^n = \Delta x^{-1} \int_{C_i} \boldsymbol{U}(x, t^n)\,\mathrm{d}x$. Applying (1.2) to $C_i \times [t^n, t^{n+1}]$ gives the *exact* relation

$$\boldsymbol{U}_i^{n+1} = \boldsymbol{U}_i^n - \frac{\Delta t}{\Delta x}(\hat{\boldsymbol{F}}_{i+1/2} - \hat{\boldsymbol{F}}_{i-1/2}), \qquad \hat{\boldsymbol{F}}_{i+1/2} = \frac{1}{\Delta t}\int_{t^n}^{t^{n+1}} \boldsymbol{F}(\boldsymbol{U}(x_{i+1/2}, t))\mathrm{d}t. \tag{1.5}$$

Any numerical method of this form is conservative: the sum of $\boldsymbol{U}_i \Delta x$ changes only through the boundary fluxes. Godunov (1959) closed (1.5) by replacing the unknown $\boldsymbol{U}(x, t^n)$ with the piecewise-constant function equal to $\boldsymbol{U}_i^n$ in C_i. Every interface then carries a Riemann problem with data $(\boldsymbol{U}_i^n, \boldsymbol{U}_{i+1}^n)$. As long as waves from neighbouring interfaces do not reach the interface during the step, i.e. under the Courant–Friedrichs–Lewy condition

$$\frac{\Delta t}{\Delta x} \max|\lambda| \le 1, \tag{1.6}$$

the solution *at* $x_{i+1/2}$ is the constant $\boldsymbol{U}^*_{i+1/2} = \boldsymbol{U}_{\mathrm{RP}}(\xi = 0; \boldsymbol{U}_i^n, \boldsymbol{U}_{i+1}^n)$, and the flux integral is exact:

$$\hat{\boldsymbol{F}}_{i+1/2} = \boldsymbol{F}(\boldsymbol{U}_{\mathrm{RP}}(0; \boldsymbol{U}_i^n, \boldsymbol{U}_{i+1}^n)). \tag{1.7}$$

(The stricter bound 1/2 guarantees that waves from neighbouring interfaces do not interact inside a cell; it is needed for the reconstruct–evolve–average interpretation but not for (1.7), which only uses the value on the interface.)

Godunov's scheme is first-order accurate, conservative, and *upwind*: it uses characteristic information to decide which side of the interface each piece of information comes from. With the exact Riemann solution it inherits the admissibility of the exact waves. Its accuracy is raised by *reconstructing* a piecewise polynomial from the averages and solving Riemann problems between the reconstructed interface values (Chapter 10). If a conservative scheme converges as the grid is refined, its limit is a weak solution (see the Lax–Wendroff theorem in LeVeque, 2002); which weak solution is a separate question, and one that will return in Chapter 8.

Why the Riemann problem is the building block

Three properties make the Riemann problem the natural "atom" of a shock-capturing scheme. (i) It contains, exactly, the nonlinear wave structure that a scheme has to respect at a discontinuity, including the correct jump conditions. (ii) Its solution is self-similar, so the flux

at $\xi = 0$ is a function of two states only, $\hat{\boldsymbol{F}} = \hat{\boldsymbol{F}}(\boldsymbol{U}_L, \boldsymbol{U}_R)$, which can be evaluated face by face. (iii) Every approximate solver used in practice—HLL, HLLC, HLLD, Roe-type—is a model of the exact Riemann fan that keeps some of its waves and discards others (Chapter 9); the exact solution is the end point against which they are measured (Toro, 2009).

1.3 Why relativistic MHD

Relativistic magnetised outflows and plasmas are central to high-energy astrophysics: jets from active galactic nuclei and X-ray binaries, pulsar winds and their nebulae, gamma-ray-burst outflows, and accretion flows and magnetospheres of compact objects. Numerical models of these systems solve the equations of relativistic MHD with Godunov-type schemes; reviews of the methods are given by Martí and Müller (2015) and in the textbook of Rezzolla and Zanotti (2013). Three features make the relativistic magnetised case harder than Newtonian MHD:

- the characteristic speeds are bounded by c and crowd together as the flow or the field becomes relativistic, and they are roots of a quartic rather than of a biquadratic;
- the conserved and primitive variables are related implicitly, so every flux evaluation needs a nonlinear inversion (Section 2.6);
- in strongly magnetised regions the gas pressure is a small difference of large numbers, $p = P - b^2/2$, so small relative errors in the total pressure produce large relative errors (or negative values) in p.

1.4 The exact solver in practice, and a roadmap

The exact solution of the SRMHD Riemann problem was given by Giacomazzo and Rezzolla (2006), building on the special case of purely tangential fields by Romero et al. (2005) and on the relativistic hydrodynamic solvers of Martí and Müller (1994), Pons et al. (2000), Rezzolla and Zanotti (2001) and Rezzolla et al. (2003). Approximate solvers of the HLL family (Mignone and Bodo, 2006; Mignone et al., 2009) and linearised solvers (Komissarov, 1999; Balsara, 2001; Antón et al., 2010) are what multidimensional codes use. These notes are written for a reader who works on an exact SRMHD solver that is used as the flux function of a two-dimensional constrained-transport code. That implementation is described where it illustrates the theory (Sections 7.4 and 8.5), always as recorded in its own documentation.

Roadmap. Chapter 2 derives the equations of ideal SRMHD and their conservative form. Chapter 3 derives the characteristic speeds, their ordering and their degeneracies. Chapters 4 and 5 treat the two kinds of elementary waves, discontinuities and rarefactions. Chapter 6 solves the relativistic hydrodynamic Riemann problem completely, as a warm-up. Chapter 7 assembles the seven-wave SRMHD problem and describes exact solution methods, including the reference implementation. Chapter 8 discusses compound waves and non-uniqueness. Chapter 9 derives the approximate solvers HLL, HLLC and HLLD and reviews linearised solvers. Chapter 10 places the Riemann solver in a multidimensional finite-volume code with constrained transport and lists the standard tests. The appendices collect the algebra of b^μ, the eigenstructure, a notation table and a record of what was verified and how.

Exercises

Exercise 1.1. Show that if $\boldsymbol{U}(x,t)$ solves (1.1) with data (1.3), then so does $\boldsymbol{U}_\alpha(x,t) = \boldsymbol{U}(\alpha x, \alpha t)$ for every $\alpha > 0$. Conclude that a unique solution is a function of x/t.

Solution. $\partial_t \boldsymbol{U}_\alpha = \alpha(\partial_t \boldsymbol{U})(\alpha x, \alpha t)$ and $\partial_x \boldsymbol{F}(\boldsymbol{U}_\alpha) = \alpha(\partial_x \boldsymbol{F}(\boldsymbol{U}))(\alpha x, \alpha t)$, so (1.1) holds; the data are scale invariant. Uniqueness gives $\boldsymbol{U} = \boldsymbol{U}_\alpha$ for all α; choose $\alpha = 1/t$.

Exercise 1.2. For linear advection $\partial_t u + a\partial_x u = 0$ with $a > 0$, show that Godunov's method is the first-order upwind scheme $u_i^{n+1} = u_i^n - a\frac{\Delta t}{\Delta x}(u_i^n - u_{i-1}^n)$, and that it is stable for $a\Delta t/\Delta x \le 1$.

Exercise 1.3. Write the integral form (1.2) on a rectangle that contains a single discontinuity moving at constant speed V_s, and let the rectangle shrink onto the discontinuity. Derive the Rankine–Hugoniot condition $\boldsymbol{F}_a - \boldsymbol{F}_b = V_s(\boldsymbol{U}_a - \boldsymbol{U}_b)$. (This is used throughout Chapter 4.)

Chapter 2

The equations of ideal SRMHD

2.1 Fluid

The rest-mass current and the energy–momentum tensor of a perfect fluid are

$$J^\mu = \rho u^\mu, \qquad T^{\mu\nu}_{\rm fl} = \rho h\, u^\mu u^\nu + p\, g^{\mu\nu}, \tag{2.1}$$

with $u^\mu u_\mu = -1$ and $h = 1 + \epsilon + p/\rho$. In the fluid rest frame $T^{00}_{\rm fl} = e \equiv \rho(1+\epsilon)$ is the energy density and $T^{ij}_{\rm fl} = p\,\delta^{ij}$.

2.2 Electromagnetic field and the ideal-MHD condition

Let $F^{\mu\nu}$ be the Faraday tensor and ${}^*F^{\mu\nu}$ its dual. In the laboratory frame we choose the components

$${}^*F^{0i} = B^i, \qquad {}^*F^{ij} = -\varepsilon^{ijk} E_k, \tag{2.2}$$

with ε^{ijk} the three-dimensional Levi-Civita symbol. With this choice the homogeneous Maxwell equations $\partial_\mu {}^*F^{\mu\nu} = 0$ read $\nabla \cdot \boldsymbol{B} = 0$ ($\nu = 0$) and $\partial_t \boldsymbol{B} + \nabla \times \boldsymbol{E} = 0$ ($\nu = j$).

The comoving (fluid-frame) electric and magnetic fields are the four-vectors

$$e^\mu = F^{\mu\nu} u_\nu, \qquad b^\mu = {}^*F^{\mu\nu} u_\nu. \tag{2.3}$$

Both are orthogonal to u^μ by antisymmetry. A perfect conductor cannot sustain an electric field in its rest frame; the *ideal-MHD condition* is $e^\mu = 0$, which in the laboratory frame reads

$$\boldsymbol{E} = -\boldsymbol{v} \times \boldsymbol{B}. \tag{2.4}$$

Proposition 2.1 (Comoving field)**.** *Under* (2.4)*,*

$$b^0 = W(\boldsymbol{v}\cdot\boldsymbol{B}) = W\eta, \tag{2.5}$$

$$b^i = \frac{B^i}{W} + b^0 v^i, \tag{2.6}$$

$$b^2 = b^\mu b_\mu = \frac{B^2}{W^2} + (\boldsymbol{v}\cdot\boldsymbol{B})^2, \tag{2.7}$$

$u_\mu b^\mu = 0$*, and* ${}^*F^{\mu\nu} = u^\mu b^\nu - u^\nu b^\mu$*.*

Proof. With $u_\nu = W(-1, \boldsymbol{v})$, (2.2) gives $b^0 = {}^*F^{0i}u_i = W\,\boldsymbol{B}\cdot\boldsymbol{v}$ and $b^i = {}^*F^{i0}u_0 + {}^*F^{ij}u_j = WB^i - W\varepsilon^{ijk}v_j E_k = W\,[\boldsymbol{B} - \boldsymbol{v}\times\boldsymbol{E}]^i$. With $\boldsymbol{E} = -\boldsymbol{v}\times\boldsymbol{B}$, $\boldsymbol{v}\times\boldsymbol{E} = -\boldsymbol{v}\times(\boldsymbol{v}\times\boldsymbol{B}) = v^2\boldsymbol{B} - \boldsymbol{v}(\boldsymbol{v}\cdot\boldsymbol{B})$, so $b^i = W[(1-v^2)B^i + v^i\eta] = B^i/W + W\eta\, v^i$, which is (2.6). Then $u_\mu b^\mu = W(-b^0 + \boldsymbol{v}\cdot\boldsymbol{b}) = W(-W\eta + \eta/W + W\eta v^2) = 0$. Finally $b^2 = -(b^0)^2 + |\boldsymbol{b}|^2 = -W^2\eta^2 + B^2/W^2 + 2\eta^2 + W^2\eta^2 v^2 = B^2/W^2 + \eta^2$. For the last statement compute $u^0 b^i - u^i b^0 = W(B^i/W + W\eta v^i) - Wv^i W\eta = B^i = {}^*F^{0i}$; the spatial components follow in the same way. □

Inverting (2.6), the laboratory field is $B^i = u^0 b^i - u^i b^0 = W(b^i - b^0 v^i)$. In the fluid rest frame $b^\mu = (0, \boldsymbol{B}')$: b^μ *is* the rest-frame magnetic field, and b^2 its invariant strength. The *magnetic pressure* is $b^2/2$.

Electromagnetic stress–energy. In our units $T^{\mu\nu}_{\rm em} = F^{\mu\alpha}F^{\nu}{}_{\alpha} - \frac{1}{4}g^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}$. Under the ideal-MHD condition,

$$T^{\mu\nu}_{\rm em} = b^2 u^\mu u^\nu + \tfrac{1}{2} b^2 g^{\mu\nu} - b^\mu b^\nu. \tag{2.8}$$

Proof. In the fluid rest frame the field is purely magnetic, $\boldsymbol{B}' = \boldsymbol{b}$, and the Maxwell tensor has $T^{00} = b^2/2$, $T^{0i} = 0$, $T^{ij} = \frac{1}{2}b^2\delta^{ij} - b^i b^j$. The right-hand side of (2.8) evaluated with $u^\mu = (1, \mathbf{0})$ and $b^\mu = (0, \boldsymbol{b})$ gives the same components. Both sides are tensors, so they agree in every frame. □

2.3 Total energy–momentum tensor and the conservation laws

Adding fluid and field,

$$\boxed{T^{\mu\nu} = (\rho h + b^2)\, u^\mu u^\nu + \Big(p + \frac{b^2}{2}\Big) g^{\mu\nu} - b^\mu b^\nu = w\, u^\mu u^\nu + P\, g^{\mu\nu} - b^\mu b^\nu} \tag{2.9}$$

The equations of ideal SRMHD are conservation of rest mass, of energy–momentum, and the homogeneous Maxwell equations:

$$\nabla_\mu(\rho u^\mu) = 0, \qquad \nabla_\mu T^{\mu\nu} = 0, \qquad \nabla_\mu {}^*F^{\mu\nu} = \nabla_\mu(u^\mu b^\nu - u^\nu b^\mu) = 0. \tag{2.10}$$

The inhomogeneous Maxwell equations only determine the current and charge densities, which ideal MHD does not need. Together with an equation of state the system (2.10) is closed.

2.4 Conservative form

In flat spacetime and Cartesian coordinates, (2.10) is a system of conservation laws $\partial_t \boldsymbol{U} + \partial_i \boldsymbol{F}^i = 0$. Reading off the time components:

Mass. $D \equiv \rho u^0 = \rho W$ with flux $F^i(D) = \rho W v^i$.

Momentum. $S_j \equiv T^{0j} = wW^2 v_j - b^0 b_j$. Substituting (2.5)–(2.7),

$$wW^2 v_j - b^0 b_j = \Big(\rho h + \frac{B^2}{W^2} + \eta^2\Big) W^2 v_j - W\eta\Big(\frac{B_j}{W} + W\eta v_j\Big) = (\rho h W^2 + B^2) v_j - \eta B_j. \tag{2.11}$$

Energy. $T^{00} = wW^2 - P - (b^0)^2$. Since $b^2W^2 - (b^0)^2 = B^2$ by (2.5)–(2.7), $T^{00} = \rho h W^2 + B^2 - p - \frac{1}{2}b^2$. One evolves either $E \equiv T^{00}$ or $\tau \equiv E - D$, which removes the rest-mass energy and is better conditioned in the Newtonian regime.

Magnetic field. The $\nu = j$ components of $\partial_\mu(u^\mu b^\nu - u^\nu b^\mu) = 0$ give $\partial_t B^j + \partial_i(u^i b^j - u^j b^i) = 0$, and $u^i b^j - u^j b^i = W v^i (B^j/W + b^0 v^j) - W v^j (B^i/W + b^0 v^i) = v^i B^j - v^j B^i$. The $\nu = 0$ component is the constraint $\partial_i B^i = 0$.

Conserved variables and x-fluxes

$$\boldsymbol{U} = \begin{pmatrix} D \\ S_j \\ \tau \\ B^k \end{pmatrix} = \begin{pmatrix} \rho W \\ (\rho h W^2 + B^2) v_j - (\boldsymbol{v}\cdot\boldsymbol{B}) B_j \\ \rho h W^2 + B^2 - p - \frac{1}{2}[B^2/W^2 + (\boldsymbol{v}\cdot\boldsymbol{B})^2] - \rho W \\ B^k \end{pmatrix}, \tag{2.12}$$

$$\boldsymbol{F}^x = \begin{pmatrix} D v_x \\ w W^2 v_x v_j + P\delta_{xj} - b^x b_j \\ S_x - D v_x \\ v_x B^k - v^k B^x \end{pmatrix}. \tag{2.13}$$

Equivalent forms: $\boldsymbol{S} = \rho h W^2 \boldsymbol{v} + \boldsymbol{E}\times\boldsymbol{B}$, $\tau = \rho h W^2 - p - D + \frac{1}{2}(E^2 + B^2)$ with $E^2 = |\boldsymbol{v}\times\boldsymbol{B}|^2 = v^2B^2 - \eta^2$, and $T^{xj} = S_j v_x + P\delta_{xj} - B^x b_j/W$.

The equivalent forms follow from $\boldsymbol{E}\times\boldsymbol{B} = -(\boldsymbol{v}\times\boldsymbol{B})\times\boldsymbol{B} = \boldsymbol{v}B^2 - \boldsymbol{B}\eta$, from $B^2 - \frac{1}{2}b^2 = \frac{1}{2}(B^2 + v^2B^2 - \eta^2)$, and from expanding $b^x b_j$ with (2.6). All of (2.12) was checked symbolically against the components of (2.9) (Chapter D).

In the code: conserved variables and fluxes

`HLLD/src/physics/hlld.py` builds $\boldsymbol{U}$ and $\boldsymbol{F}$ exactly as in (2.12): `primitive_to_conserved` uses $\tau = \rho h W^2 - p - D + \frac{1}{2}(B^2 + B^2v^2 - \eta^2)$, and `compute_srmhd_fluxes` uses $\tau = (\rho h + b^2)W^2 - (p + b^2/2) - (b^0)^2 - D$, the momentum flux $wWv_xu_j + P\delta_{xj} - b^x b_j$ and the induction flux $v_xB^k - v^kB^x$, for any sweep direction (`idir`). The HLLD star-state algebra is written for the total energy, so `energy_form` converts $\tau \to E = \tau + D$ and $F_\tau \to F_E = F_\tau + F_D$. The exact solver (`rmhd_final/rmhd/batched/jump.py`, `cons_flux`) also uses $E = T^{00}$; its state vector is $(\rho, P, v_x, v_y, v_z, B_y, B_z)$ with $B_n = B_x$ held separately, and it carries the *total* pressure P, not p.

One dimension. In planar symmetry $\partial_y = \partial_z = 0$, the constraint gives $\partial_x B^x = 0$ and the x-component of the induction equation gives $\partial_t B^x = 0$: the normal field is a constant parameter of the Riemann problem. The one-dimensional system has seven unknowns, $\boldsymbol{U} = (D, S_x, S_y, S_z, \tau, B^y, B^z)$.

2.5 Equation of state and sound speed

We use the ideal gas $p = (\Gamma - 1)\rho\epsilon$, so

$$h = 1 + \frac{\Gamma}{\Gamma - 1}\frac{p}{\rho}. \tag{2.14}$$

The relativistic sound speed is $c_s^2 = (\partial p/\partial e)_s$ with $e = \rho(1+\epsilon)$ (Antón et al., 2010). Along an isentrope the first law gives $\mathrm{d}\epsilon = (p/\rho^2)\mathrm{d}\rho$, hence $\mathrm{d}e = (1+\epsilon)\mathrm{d}\rho + \rho\,\mathrm{d}\epsilon = h\,\mathrm{d}\rho$ and

$$c_s^2 = \frac{1}{h}\Big(\frac{\partial p}{\partial \rho}\Big)_s = \frac{\Gamma p}{\rho h} = \frac{\Gamma(\Gamma-1)p}{(\Gamma-1)\rho + \Gamma p} < \Gamma - 1, \tag{2.15}$$

the bound being approached as $p/\rho \to \infty$; causality ($c_s < 1$) holds for $\Gamma \le 2$.

In the code: equation of state

`HLLD/src/physics/eos.py` implements a hybrid EOS, $p = K\rho^{\Gamma} + (\Gamma_{\rm th} - 1)\rho(\epsilon - \epsilon_{\rm cold})$, with $c_s^2 = (\Gamma p_{\rm cold} + \Gamma_{\rm th} p_{\rm th})/(\rho h)$. All production runs and scripts construct it with $K = 0$ and $\Gamma_{\rm th} = \Gamma$, which is the ideal gas above. The exact solver forms $p = P - b^2/2$ inside every routine and uses $h = 1 + \Gamma p/[(\Gamma - 1)\rho]$ and $c_s^2 = \Gamma p/(\rho h)$ (`rmhd/batched/wave_speeds_b.py`).

2.6 Primitive recovery

The fluxes are functions of the primitive variables $(\rho, \boldsymbol{v}, p, \boldsymbol{B})$, but the scheme evolves $\boldsymbol{U}$. Recovering the primitives is a nonlinear problem with no closed-form solution. Its structure is easiest to see with the variable $Z \equiv \rho h W^2$. From (2.12),

$$\boldsymbol{S} \cdot \boldsymbol{B} = (\rho h W^2 + B^2)\eta - \eta B^2 = Z\,\eta, \tag{2.16}$$

so $\eta = (\boldsymbol{S} \cdot \boldsymbol{B})/Z$; squaring $\boldsymbol{S}$ then gives

$$v^2 = \frac{S^2 Z^2 + (\boldsymbol{S} \cdot \boldsymbol{B})^2 (2Z + B^2)}{Z^2 (Z + B^2)^2}, \tag{2.17}$$

and the energy equation becomes one equation in Z (Mignone and Bodo, 2006),

$$f(Z) \equiv Z - p(Z) + \Big(1 - \frac{1}{2W^2}\Big) B^2 - \frac{(\boldsymbol{S} \cdot \boldsymbol{B})^2}{2Z^2} - E = 0, \qquad p(Z) = \frac{\Gamma - 1}{\Gamma}\,\frac{Z - DW}{W^2}, \tag{2.18}$$

where $W = W(Z)$ through (2.17) and $E = \tau + D$. The identities (2.17)–(2.18) were checked in Chapter D. Noble et al. (2006) compared one- and two-variable schemes of this kind for general relativistic MHD; Kastaun et al. (2021) reformulated the inversion as a root-finding problem in $\mu = 1/(\rho h W)$ with a bracketing interval that is guaranteed to contain the unique solution, which makes it robust across the physical parameter space.

In the code: primitive recovery

`HLLD/src/physics/c2p.py` implements the scheme of Kastaun et al. (2021), vectorised over cells: a bracketed root-find for $\mu = 1/(\rho h W)$. According to the guide (Part 2.6) the bracketed secant was found to stall on hot, fast gas (e.g. $(\rho, p, v) = (0.1, 1, 0.99)$ left a relative error $\sim 10^{-6}$), so cells that it leaves unconverged are re-solved by bisection; cells where recovery still fails are counted (`c2p_bad`) and floored.

2.7 Newtonian limit

Introduce a bookkeeping parameter δ with $\boldsymbol{v} = O(\delta)$, $\boldsymbol{B} = O(\delta)$ (Alfvén speeds comparable with flow speeds) and $p, \rho\epsilon = O(\delta^2)\rho$. Expanding (2.12) (checked symbolically, Chapter D):

$$D = \rho + O(\delta^2), \qquad S_x = \rho v_x + O(\delta^3), \tag{2.19}$$

$$\tau = \rho\epsilon + \tfrac{1}{2}\rho v^2 + \tfrac{1}{2}B^2 + O(\delta^4), \qquad T^{xx} = \rho v_x^2 + p + \tfrac{1}{2}B^2 - B_x^2 + O(\delta^4), \tag{2.20}$$

and the energy flux $S_x - D v_x = (\tfrac{1}{2}\rho v^2 + \rho\epsilon + p + B^2) v_x - (\boldsymbol{v} \cdot \boldsymbol{B}) B_x + O(\delta^5)$. These are the conserved variables and fluxes of Newtonian ideal MHD in Heaviside–Lorentz units; the induction equation is unchanged. The relativistic corrections enter through W, through the enthalpy ρh as inertia, and through the electric-field terms $\boldsymbol{E} \times \boldsymbol{B}$ and $E^2/2$ (displacement current), which make the momentum of the field $\boldsymbol{v}B^2 - \boldsymbol{B}\eta$ part of $\boldsymbol{S}$.

Exercises

Exercise 2.1. Verify $u_\mu b^\mu = 0$ and $b^2 = B^2/W^2 + \eta^2$ directly from (2.5)–(2.6), and show that $b^2W^2 - (b^0)^2 = B^2$.

Solution. $b^2W^2 - (b^0)^2 = (B^2/W^2 + \eta^2)W^2 - W^2\eta^2 = B^2$.

Exercise 2.2. Show that $\boldsymbol{S} = \rho h W^2 \boldsymbol{v} + \boldsymbol{E} \times \boldsymbol{B}$ and $\tau = \rho h W^2 - p - D + \frac{1}{2}(E^2 + B^2)$, and interpret each term.

Exercise 2.3. Show that $T^{ij} = S_j v^i + P\delta^{ij} - B^i b^j / W$. (Expand $b^i b^j$ with (2.6).)

Exercise 2.4. Show that $c_s^2 < \Gamma - 1$ for the ideal gas and that $c_s \to 0$ as $p/\rho \to 0$. For $\Gamma = 4/3$ what is the largest possible sound speed?

Solution. From (2.15), $c_s^2 = (\Gamma - 1)\Gamma p/[(\Gamma - 1)\rho + \Gamma p]$, increasing in p/ρ with limit $\Gamma - 1$; for $\Gamma = 4/3$, $c_s < 1/\sqrt{3}$.

Exercise 2.5. Derive (2.17) and (2.18). Show that for $\boldsymbol{B} = 0$ the problem reduces to one equation for Z with $v^2 = S^2/(Z^2)$, i.e. $v = S/(E + p)$.

Exercise 2.6. Expand τ to fourth order in δ and identify the leading relativistic correction to the kinetic energy.

Solution. With $W^2 = 1 + v^2 + v^4 + \dots$ and $D = \rho W$, $\rho h W^2 - p - D = \rho(1 + \epsilon)W^2 + p(W^2 - 1) - \rho W$; the kinetic part $\rho(W^2 - W) = \rho(\frac{1}{2}v^2 + \frac{5}{8}v^4 + \dots)$, so the first correction is $\frac{5}{8}\rho v^4$, plus $\rho\epsilon v^2 + pv^2$ from the internal energy and pressure.

Chapter 3

Characteristics and hyperbolicity

3.1 The one-dimensional system

In planar symmetry the system (2.12) has seven components, $\boldsymbol{U} = (D, S_x, S_y, S_z, \tau, B^y, B^z)^{\mathsf{T}}$, and B^x is a constant parameter. Written in quasi-linear form,

$$\partial_t \boldsymbol{U} + \mathsf{A}(\boldsymbol{U})\,\partial_x \boldsymbol{U} = 0, \qquad \mathsf{A} = \frac{\partial \boldsymbol{F}^x}{\partial \boldsymbol{U}}, \tag{3.1}$$

or, in primitive variables $\boldsymbol{V} = (\rho, v_x, v_y, v_z, p, B^y, B^z)$,

$$\partial_t \boldsymbol{V} + \tilde{\mathsf{A}}\,\partial_x \boldsymbol{V} = 0, \qquad \tilde{\mathsf{A}} = \Big(\frac{\partial \boldsymbol{U}}{\partial \boldsymbol{V}}\Big)^{-1} \frac{\partial \boldsymbol{F}^x}{\partial \boldsymbol{V}}. \tag{3.2}$$

The two matrices are similar and have the same eigenvalues, the *characteristic speeds* λ. A small-amplitude disturbance proportional to a right eigenvector $\boldsymbol{r}_k$ travels at λ_k. Writing A out explicitly is laborious; it is much more economical to find the characteristic speeds covariantly and then specialise to the x-direction, which is what we do now.

3.2 Characteristic speeds from a covariant plane-wave analysis

A characteristic hypersurface $\phi(x^\mu) = \text{const}$ is a surface across which derivatives of the solution may jump. For a plane front moving along x with laboratory speed λ we take $\phi = x - \lambda t$, i.e.

$$\phi_\mu \equiv \partial_\mu \phi = (-\lambda, 1, 0, 0). \tag{3.3}$$

Following Anile (1989), Komissarov (1999) and Antón et al. (2010) we build three Lorentz scalars from ϕ_μ:

$$a \equiv u^\mu \phi_\mu = W(v_x - \lambda), \qquad \mathcal{B} \equiv b^\mu \phi_\mu = b^x - \lambda b^0, \qquad G \equiv \phi^\mu \phi_\mu = 1 - \lambda^2. \tag{3.4}$$

(Mignone and Bodo (2006) use $a = W(\lambda - v_x)$; only a^2 enters the quartic below.)

Lemma 3.1 (Rest-frame meaning of the scalars)**.** *Let V be the phase speed of the front measured in the fluid rest frame and θ the angle between the rest-frame magnetic field $\boldsymbol{b}$ and the front normal. Then*

$$V^2 = \frac{a^2}{G + a^2}, \qquad b^2 \cos^2\theta = \frac{\mathcal{B}^2}{G + a^2}. \tag{3.5}$$

Proof. In the rest frame $u^\mu = (1, \boldsymbol{0})$ and $b^\mu = (0, \boldsymbol{b})$; write $\phi_\mu = (\phi'_0, \boldsymbol{\phi}')$. The front $\phi'_0 t' + \boldsymbol{\phi}' \cdot \boldsymbol{x}' = \text{const}$ moves along $\boldsymbol{\phi}'/|\boldsymbol{\phi}'|$ with phase speed $V = -\phi'_0/|\boldsymbol{\phi}'|$. Now $a = \phi'_0$, $G = -\phi'^2_0 + |\boldsymbol{\phi}'|^2$, so $|\boldsymbol{\phi}'|^2 = G + a^2$ and $V^2 = a^2/(G + a^2)$. Also $\mathcal{B} = \boldsymbol{b} \cdot \boldsymbol{\phi}' = b|\boldsymbol{\phi}'|\cos\theta$. Both relations involve only scalars and hold in every frame. □

3.2.1 The dispersion relation in the fluid rest frame

Linearise the equations about a uniform state at rest, $u^\mu = (1, \mathbf{0})$, $b^\mu = (0, \boldsymbol{b})$, with perturbations proportional to $\exp[ik(x - Vt)]$, and choose axes so that $\boldsymbol{b} = b(\cos\theta, \sin\theta, 0)$. To first order $\delta u^\mu = (0, \delta\boldsymbol{v})$; $u_\mu b^\mu = 0$ gives $\delta b^0 = \boldsymbol{b}\cdot\delta\boldsymbol{v}$; $B^i = u^0 b^i - u^i b^0$ gives $\delta\boldsymbol{B} = \delta\boldsymbol{b}$; and $\delta P = \delta p + \boldsymbol{b}\cdot\delta\boldsymbol{b}$. From (2.9), $\delta T^{0i} = w\,\delta v^i - (\boldsymbol{b}\cdot\delta\boldsymbol{v})\,b^i$ and $\delta T^{ij} = \delta P\,\delta^{ij} - b^i\delta b^j - \delta b^i b^j$. The divergence constraint gives $\delta b_x = 0$. The linearised equations are (after dividing by ik)

$$-V\delta\rho + \rho\,\delta v_x = 0, \tag{3.6}$$

$$-V\big[w\,\delta\boldsymbol{v} - (\boldsymbol{b}\cdot\delta\boldsymbol{v})\boldsymbol{b}\big] + \hat{\boldsymbol{x}}\,\delta P - b\cos\theta\,\delta\boldsymbol{b} = 0, \tag{3.7}$$

$$-V\delta\boldsymbol{b} = b\cos\theta\,\delta\boldsymbol{v} - \boldsymbol{b}\,\delta v_x, \tag{3.8}$$

$$\delta p = h c_s^2\,\delta\rho. \tag{3.9}$$

The term $(\boldsymbol{b}\cdot\delta\boldsymbol{v})\boldsymbol{b}$ in (3.7) is the relativistic one: it comes from $b^0 b^i$ in the momentum density, i.e. from the displacement current.

Alfvén waves. The z-components decouple: $-Vw\,\delta v_z - b\cos\theta\,\delta b_z = 0$ and $-V\delta b_z = b\cos\theta\,\delta v_z$ give

$$V^2 = c_a^2\cos^2\theta, \qquad c_a^2 \equiv \frac{b^2}{w} = \frac{b^2}{\rho h + b^2}. \tag{3.10}$$

$c_a < 1$ always, in contrast with the Newtonian Alfvén speed $B/\sqrt{\rho}$.

Magnetosonic waves. Eliminate $\delta\rho$, δp and $\delta b_y = (b/V)(\sin\theta\,\delta v_x - \cos\theta\,\delta v_y)$, so that $\delta P = V^{-1}[(\rho h c_s^2 + b^2\sin^2\theta)\delta v_x - b^2\sin\theta\cos\theta\,\delta v_y]$. The x- and y-components of (3.7), multiplied by V, become

$$\begin{pmatrix} \rho h c_s^2 + b^2\sin^2\theta - V^2(\rho h + b^2\sin^2\theta) & -(1-V^2)b^2\sin\theta\cos\theta \\ -(1-V^2)b^2\sin\theta\cos\theta & b^2\cos^2\theta - V^2(\rho h + b^2\cos^2\theta) \end{pmatrix} \begin{pmatrix} \delta v_x \\ \delta v_y \end{pmatrix} = 0. \tag{3.11}$$

Its determinant equals $\rho h\,[\,wV^4 - (b^2 + \rho h c_s^2 + c_s^2 b^2\cos^2\theta)V^2 + c_s^2 b^2\cos^2\theta\,]$ (checked symbolically), so with $c_a^2 = b^2/w$:

$$\boxed{V^4 - (c_a^2 + c_s^2 - c_s^2 c_a^2\sin^2\theta)V^2 + c_s^2 c_a^2\cos^2\theta = 0} \tag{3.12}$$

The Newtonian relation has $c_a^2 + c_s^2$ in the middle coefficient; the relativistic correction $-c_s^2 c_a^2\sin^2\theta$ comes from the displacement-current term. Together with the entropy mode ($V = 0$, $\delta p = \delta\boldsymbol{v} = \delta\boldsymbol{b} = 0$, $\delta\rho \neq 0$), (3.10) and (3.12) account for all seven modes.

3.2.2 Laboratory-frame speeds

Substituting (3.5) into the rest-frame relations and multiplying by powers of $G + a^2$ turns them into polynomials in λ:

Entropy (contact) wave. $a = 0$:

$$\lambda_{\rm e} = v_x. \tag{3.13}$$

Alfvén waves. $V^2 = c_a^2\cos^2\theta$ becomes $w\,a^2 = \mathcal{B}^2$, i.e. $\sqrt{w}\,W(v_x - \lambda) = \pm(b^x - \lambda b^0)$. Solving for λ,

$$\boxed{\lambda_A^{(\pm)} = \frac{b^x \pm \sqrt{w}\,u^x}{b^0 \pm \sqrt{w}\,u^0} = v_x + \frac{B^x(1-v^2)}{\eta \pm \sqrt{w}}} \tag{3.14}$$

The second form follows by inserting (2.5)–(2.6) and dividing numerator and denominator by W. It is the form used by Giacomazzo and Rezzolla (2006) and by the code.

Proposition 3.1 (Which Alfvén speed is which)**.** $\sqrt{w} > |\eta|$, *hence* $\operatorname{sign}(\lambda_A^{(\pm)} - v_x) = \pm\operatorname{sign}(B^x)$. *For* $B^x > 0$, $\lambda_A^{(-)} < v_x < \lambda_A^{(+)}$; *for* $B^x < 0$ *the labels are reversed.*

Proof. $w = \rho h + b^2 > b^2 = B^2/W^2 + \eta^2 \geq \eta^2$, so $\eta + \sqrt{w} > 0 > \eta - \sqrt{w}$; the sign of $\lambda_A^{(\pm)} - v_x$ is that of $B^x/(\eta \pm \sqrt{w})$. □

In the code: the Alfvén pair for $B_n < 0$

The labels $(\pm)$ in (3.14) refer to the sign in front of $\sqrt{w}$, not to the direction of propagation; Antón et al. (2010) make the same remark about their eq. (38). In the exact solver every helper that returned the pair as `(la, ra)` (left, right) had used the $(-)$ root as "left". For $B_n < 0$ this built rotations of the wrong family. The code now orders the pair by value (e.g. `rmhd/shock_speed.py`: `if not (la <= ra): la, ra = ra, la`, commented "labelled by the sign of sqrt(rho h+b^2): swapped for B_n < 0").

Magnetosonic waves. Inserting (3.5) into (3.12), multiplying by $(G + a^2)^2$ and using $c_a^2/b^2 = 1/w$ gives $(1 - \zeta^2)a^4 - G\,(\zeta^2 a^2 - c_s^2\mathcal{B}^2/w) = 0$ with $\zeta^2 = c_s^2 + c_a^2 - c_s^2 c_a^2$. Multiplying by w and using $w(1 - \zeta^2) = \rho h(1 - c_s^2)$ and $w\zeta^2 = b^2 + \rho h c_s^2$:

$$\boxed{\rho h(1 - c_s^2)\,a^4 - (1 - \lambda^2)\big[(b^2 + \rho h c_s^2)\,a^2 - c_s^2\mathcal{B}^2\big] = 0, \qquad a = W(v_x - \lambda), \quad \mathcal{B} = b^x - \lambda b^0} \tag{3.15}$$

a quartic in λ whose four roots are the slow and fast speeds. We checked (Chapter D) that (3.15) is identical to eq. (56) of Mignone and Bodo (2006), to the normalised form eq. (4.50) of Giacomazzo and Rezzolla (2006) and eq. (16) of Del Zanna et al. (2003) (which divide by w and use $\tilde{b}^\mu = b^\mu/\sqrt{w}$), to $c_s^2\mathcal{N}_4 = 0$ with $\mathcal{N}_4$ of Antón et al. (2010, eq. 35), and coefficient by coefficient to the quartic coded in `rmhd/wave_speeds.py` and `rmhd/batched/wave_speeds_b.py`.[1]

Remark 3.1 (The full characteristic determinant). In the covariant variables (u^μ, b^μ, p, s) of Anile (1989) the system has ten unknowns, and the characteristic determinant factorises as $\det(\mathcal{A}^\alpha\phi_\alpha) = w\,a^2\mathcal{A}^2\mathcal{N}_4$ with $\mathcal{A} = wa^2 - \mathcal{B}^2$ (Antón et al., 2010, eq. 33). The squares reflect duplicate, unphysical roots introduced by the constraints $u_\mu u^\mu = -1$ and $u_\mu b^\mu = 0$; the physical speeds are the seven roots of $a\,\mathcal{A}\mathcal{N}_4 = 0$ found above. Chapter B gives the matrix and the eigenvectors.

3.3 Rest-frame speeds and their ordering

Write $X = V^2$ and $P(X) = X^2 - QX + c_s^2 c_a^2\cos^2\theta$ with $Q = c_a^2 + c_s^2 - c_s^2 c_a^2\sin^2\theta$. Let $X_s \leq X_f$ be the roots (slow and fast).

Proposition 3.2 (Ordering and bounds)**.** *For* $0 < c_s < 1$ *and* $0 \leq c_a < 1$,

$$0 \leq X_s \leq c_a^2\cos^2\theta \leq X_f \leq \zeta^2 < 1, \qquad \zeta^2 \leq 2X_f. \tag{3.16}$$

Equality in any of the first three inequalities requires $\cos\theta = 0$ *or* $\sin\theta = 0$.

Proof. P is an upward parabola in X.

(i) $P(0) = c_s^2 c_a^2\cos^2\theta \geq 0$ and $P(c_a^2\cos^2\theta) = -c_a^4\cos^2\theta\sin^2\theta\,(1 - c_s^2) \leq 0$, so the Alfvén value lies between the roots and both roots are non-negative.

(ii) $Q - \zeta^2 = c_s^2 c_a^2\cos^2\theta \geq 0$, hence $P(\zeta^2) = c_s^2 c_a^2\cos^2\theta\,(1 - \zeta^2) \geq 0$; and the vertex lies to the left of ζ^2, because $\zeta^2 - Q/2 = \frac{1}{2}(\zeta^2 - c_s^2 c_a^2\cos^2\theta) \geq 0$. Therefore ζ^2 lies to the right of both roots, $X_f \leq \zeta^2$.

[1] The same quartic is attributed also to Leismann et al. (2005); that paper's full text could not be accessed for this check, so the comparison with it was not made.

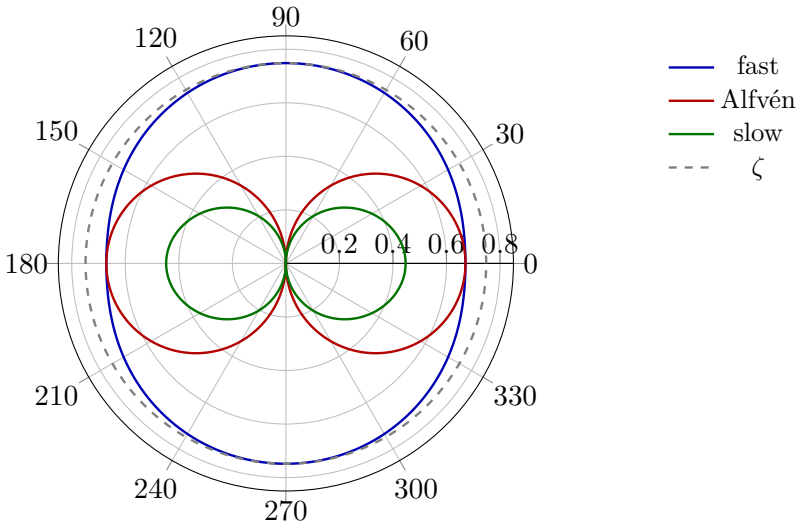


Figure 3.1: Rest-frame phase speeds $V(\theta)$ for $c_s^2 = 0.2$, $c_a^2 = 0.45$ (θ measured from $\boldsymbol{b}$, which points to the right). The fast speed reaches $\zeta = \sqrt{0.56}$ (dashed) at $\theta = 90°$; at $\theta = 0$ the Alfvén speed coincides with the fast speed (a Type II degeneracy, Section 3.5).

(iii) $\zeta^2 = 1 - (1 - c_s^2)(1 - c_a^2) < 1$.
(iv) $X_f \geq Q/2 \geq \zeta^2/2$.
The polynomial identities used here were checked symbolically. □

So in the fluid frame the speeds are ordered exactly as in Newtonian MHD, slow ≤ Alfvén ≤ fast, all are subluminal, and the fast speed is largest perpendicular to the field, where it equals ζ. Figure 3.1 shows the phase-speed (Friedrichs) diagram.

3.4 Properties of the laboratory-frame speeds

Ordering. The seven laboratory speeds are ordered as

$$\lambda_{\mathrm{f}}^{-} \leq \lambda_{\mathrm{A}}^{-} \leq \lambda_{\mathrm{s}}^{-} \leq \lambda_{\mathrm{e}} \leq \lambda_{\mathrm{s}}^{+} \leq \lambda_{\mathrm{A}}^{+} \leq \lambda_{\mathrm{f}}^{+}, \tag{3.17}$$

where now the superscripts denote the lower and upper value of each pair (Antón et al., 2010, eq. 42), a result due to Anile (1989). Aberration destroys the symmetry of each pair about v_x. We do not reprove (3.17); the following lemma is the key monotonicity behind it and is useful in its own right.

Lemma 3.2 (Monotone map)**.** *Along the x-direction, the rest-frame phase speed $V(\lambda)$ of (3.5) increases monotonically from 0 at $\lambda = v_x$ to 1 at $\lambda = \pm 1$ on either side of v_x.*

Proof. $V^2 = 1/(1 + R)$ with $R = G/a^2 = (1 - \lambda^2)/[W^2(v_x - \lambda)^2]$, and $\mathrm{d}R/\mathrm{d}\lambda = 2(1 - \lambda v_x)/[W^2(v_x - \lambda)^3]$, which is negative for $\lambda > v_x$ and positive for $\lambda < v_x$ (since $1 - \lambda v_x > 0$). □

Proposition 3.3 (An isotropic bound on the fast speeds)**.** *Let $\lambda_\zeta^\pm$ be the roots of $(1 - \zeta^2)W^2(\lambda - v_x)^2 = \zeta^2(1 - \lambda^2)$,*

$$\lambda_\zeta^\pm = \frac{v_x(1 - \zeta^2) \pm \zeta\sqrt{(1 - v^2)\,[1 - v^2\zeta^2 - v_x^2(1 - \zeta^2)]}}{1 - v^2\zeta^2}. \tag{3.18}$$

Then $\lambda_\zeta^- \leq \lambda_{\mathrm{f}}^-$ and $\lambda_{\mathrm{f}}^+ \leq \lambda_\zeta^+$.

Proof. At $\lambda_\zeta^\pm$ the rest-frame phase speed is exactly ζ; at $\lambda_{\rm f}^\pm$ it is $\sqrt{X_f(\theta)} \le \zeta$ (Proposition 3.2). By Lemma 3.2 the fast root lies closer to v_x. □

Equation (3.18) is the speed of a hypothetical isotropic wave of rest-frame speed ζ, boosted with the flow. Gammie et al. (2003) introduced this replacement of the full dispersion relation (their eq. 28) and noted that it overestimates the comoving fast speed by a bounded factor; Proposition 3.2 makes this precise: $\zeta \le \sqrt{2}\,\sqrt{X_f}$. Because it needs only a square root it is a convenient signal-speed estimate; HARM uses it, and so does the reference code.

Special cases. For $B^x = 0$ one finds $\mathcal{B} = b^x - \lambda b^0 = \eta\, a$ and (3.15) factorises as

$$a^2\Big\{\rho h(1-c_s^2)a^2 - (1-\lambda^2)\big[\rho h c_s^2 + b^2 - c_s^2\eta^2\big]\Big\} = 0, \tag{3.19}$$

so the slow speeds collapse onto $\lambda_e = v_x$ (as do both Alfvén speeds, by (3.14)) and the fast pair solves a quadratic; this is eq. (58) of Mignone and Bodo (2006). For $\boldsymbol{v} = 0$, (3.15) reduces to the biquadratic $w\lambda^4 - (b^2 + \rho h c_s^2 + c_s^2 B_x^2)\lambda^2 + c_s^2 B_x^2 = 0$, their eq. (57), which is (3.12) with $\cos^2\theta = B_x^2/B^2$.

In the code: wave speeds

`rmhd/wave_speeds.py` (scalar) and `rmhd/batched/wave_speeds_b.py` (batched) compute the coefficients of (3.15) divided by w, with $\tilde{b}^\mu = b^\mu/\sqrt{w}$ and `eps2` $= \zeta^2$, exactly as in eq. (4.50) of Giacomazzo and Rezzolla (2006). The scalar code has a separate expression for $|B_x| \le 10^{-15}$. The batched code finds the roots by Ferrari's closed form (default) or the eigenvalues of the companion matrix, polishes them with Newton iterations on the quartic, and returns the Alfvén speeds from the same intermediates. The HLL/HLLD signal speeds in `HLLD/src/physics/hlld.py` (`compute_cp_cm`) are the isotropic bound (3.18) with `v02` $= \zeta^2$; $c_{\max}$ and $c_{\min}$ are then clamped to be non-negative.

3.5 Degeneracies

The inequalities in (3.17) can be equalities. The classification follows Komissarov (1999) and Antón et al. (2010), who characterise it by the component b_n of b^μ along the Alfvén front normal in the fluid frame, $b_n^2 = \mathcal{B}^2/(G + a^2)$ evaluated at the Alfvén speed:

Type I ($b_n = 0$; for x-propagation, $B^x = 0$): both Alfvén speeds and both slow speeds coincide with the entropy speed, $\lambda_A^\pm = \lambda_s^\pm = \lambda_e = v_x$. The Riemann fan has three waves: two fast waves and a tangential discontinuity.

Type II ($b_t = 0$, i.e. $b_n^2 = b^2$; the field is along the wave normal in the fluid frame): an Alfvén speed coincides with the fast or the slow speed of the same class. In Newtonian MHD both Alfvén waves are then degenerate. In RMHD, because of aberration, the condition can hold for one Alfvén wave and not the other; only when the tangential velocity vanishes is the classical behaviour recovered (Antón et al., 2010).

Type II′ (Type II with $c_s^2 = c_a^2$): an Alfvén speed coincides with *both* magnetosonic speeds of its class (a triple point, the relativistic counterpart of the triple umbilic point of Newtonian MHD; cf. Fig. 3.1 at $\theta = 0$ when $c_s = c_a$).

At degenerate states the eigenvectors of Chapter B vanish or become linearly dependent, so they no longer form a basis; for a solver that projects on characteristic fields this is fatal. Brio and Wu (1988) cured this in Newtonian MHD by renormalising the eigenvectors; Komissarov (1999) gave relativistic eigenvectors valid through Type II degeneracies; Antón et al. (2010) gave renormalised right and left eigenvectors that are regular and complete at all physical states, including both types (their eqs. 52 and 65–67 for the Alfvén waves).

Loss of strict hyperbolicity and of convexity. At degenerate states the system is hyperbolic but not strictly hyperbolic. Moreover, the entropy and Alfvén characteristic fields are linearly degenerate ($\nabla_{\boldsymbol{U}}\lambda_k \cdot \boldsymbol{r}_k \equiv 0$), while the magnetosonic fields are not genuinely nonlinear everywhere: their convexity depends on the magnetic field (Brio and Wu, 1988; Ibáñez et al., 2015). Both properties are hypotheses of the classical existence and uniqueness theory, and their failure is what allows compound waves and non-unique solutions (Chapter 8).

Near-degeneracy. For a numerical solver, nearly coincident speeds are worse than exactly coincident ones: the system is nominally seven-wave but the Jacobian of any formulation that treats the two waves separately is nearly singular. On the interface states of the reference 2D rotor run the Alfvén and slow speeds differ by a median of 5.5×10^{-3}, against 0.28 between fast and Alfvén (the guide, §1.2).

In the code: classifying an interface

`rmhd/batched/classify.py` classifies each interface before any solve, with thresholds relative to the width of the wave fan: `FULL7` (seven waves), `COPLANAR`, `ALFVEN_DEGEN` (Type I), `SLOW_DEGEN` and `FAST_DEGEN` (Type II with the slow or fast wave). The degenerate classes are handed to a reduced solver (the guide, §1.7).

Exercises

Exercise 3.1. Show that the Alfvén speeds (3.14) are the roots of $\mathcal{A} = wa^2 - \mathcal{B}^2 = 0$, and that $\lambda_A^{(+)}\lambda_A^{(-)}$ and $\lambda_A^{(+)} + \lambda_A^{(-)}$ are rational functions of the state. Evaluate both speeds for $\boldsymbol{v} = 0$.

Solution. For $\boldsymbol{v} = 0$: $\eta = 0$, $W = 1$, so $\lambda_A^{(\pm)} = \pm B^x/\sqrt{w} = \pm c_a \cos\theta$, the rest-frame result.

Exercise 3.2. Carry out the elimination leading to (3.12) and show that for $\theta = \pi/2$ the fast speed is ζ and the slow speed vanishes, while for $\theta = 0$ the roots are c_s^2 and c_a^2.

Exercise 3.3. Show that $\zeta \le \sqrt{2}\max(c_s, c_a)$ and find the state for which the isotropic bound overestimates the fast speed the most. Compare with the statement of Gammie et al. (2003).

Solution. The worst case is $\theta = 0$, where $X_f = \max(c_s^2, c_a^2)$; for $c_s = c_a = c \ll 1$, $\zeta^2 = 2c^2 - c^4 \approx 2X_f$.

Exercise 3.4. Starting from (3.15) with $B^x = 0$, derive (3.19). Show that if moreover $\boldsymbol{v}\cdot\boldsymbol{B} = 0$ the fast speeds are those of a gas with sound speed ζ.

Exercise 3.5. For $B^x < 0$, which of $\lambda_A^{(\pm)}$ is the left-going Alfvén speed? Construct a state with $B^x = -1$ and check numerically that using the $(-)$ root as "left" gives a speed larger than v_x.

Exercise 3.6. Using Lemma 3.2, show that every characteristic speed satisfies $|\lambda| < 1$ whenever $c_s < 1$.

Solution. All rest-frame speeds are $\le \zeta < 1$; the monotone map sends $V < 1$ to $|\lambda| < 1$.

Chapter 4

Discontinuities

4.1 Rankine–Hugoniot conditions

Let a discontinuity move with constant speed V_s and separate the state a ahead of it from the state b behind it. Applying the integral form (1.2) to a small rectangle that the discontinuity crosses diagonally and letting it shrink gives the Rankine–Hugoniot (RH) conditions

$$[\![\boldsymbol{F}^x]\!] = V_s[\![\boldsymbol{U}]\!], \qquad \text{i.e.} \quad \boldsymbol{F}^x_a - \boldsymbol{F}^x_b = V_s(\boldsymbol{U}_a - \boldsymbol{U}_b). \tag{4.1}$$

With (2.12)–(2.13) these are seven conditions (mass, three momentum components, energy, two tangential field components), plus $[\![B^x]\!] = 0$.

Covariant form. The front is the hypersurface $x - V_s t = \text{const}$, with unit spacelike normal

$$n_\mu = W_s(-V_s, 1, 0, 0), \qquad n_\mu n^\mu = 1, \qquad W_s = (1 - V_s^2)^{-1/2}. \tag{4.2}$$

For any tensor current, $X^{\mu\nu}n_\nu = W_s(X^{\mu x} - V_s X^{\mu 0})$, so (4.1) is equivalent to (Anile, 1989; Giacomazzo and Rezzolla, 2006)

$$[\![\rho u^\mu]\!] n_\mu = 0, \qquad [\![T^{\mu\nu}]\!] n_\nu = 0, \qquad [\![{}^*F^{\mu\nu}]\!] n_\nu = 0. \tag{4.3}$$

For the last one, ${}^*F^{00} = 0$ and ${}^*F^{0x} = B^x$ give $[\![B^x]\!] = 0$, while ${}^*F^{j0} = -B^j$ and ${}^*F^{jx} = u^j b^x - u^x b^j = -(v_x B^j - v^j B^x)$ give the induction jump $V_s[\![B^j]\!] = [\![v_x B^j - v^j B^x]\!]$.

4.2 Mass flux and the classification of discontinuities

Define the rest-mass flux through the front and the normal component of the comoving field,

$$J \equiv -\rho u^\mu n_\mu = \rho W W_s(V_s - v_x), \qquad B_n \equiv b^\mu n_\mu = \frac{W_s}{W} B^x - \frac{\eta J}{\rho}. \tag{4.4}$$

These are eqs. (4.5) and (4.28) of Giacomazzo and Rezzolla (2006). The first condition of (4.3) says $[\![J]\!] = 0$. It also gives a useful identity: for any Q,

$$[\![Q\,(V_s - v_x)]\!] = \frac{J}{W_s}\left[\!\!\left[\frac{Q}{D}\right]\!\!\right], \tag{4.5}$$

because $V_s - v_x = J/(W_s D)$ on both sides. Two classes follow:

- $J = 0$: no mass crosses the front; it moves with the fluid, $V_s = v_{x,a} = v_{x,b}$. These are *contact* and *tangential* discontinuities.

- $J \neq 0$: mass crosses the front. These are *shocks* (compressive, $[\![\rho]\!] \neq 0$) and *rotational* (Alfvén) discontinuities.

It is convenient to write the projections of the fluxes on n_ν with (4.4). With $u^\mu n_\mu = -J/\rho$ and $h_t \equiv w/\rho = h + b^2/\rho$ (the total specific enthalpy),

$$T^{\mu\nu} n_\nu = -J h_t \, u^\mu + P \, n^\mu - B_n \, b^\mu, \tag{4.6}$$

$$^*F^{\mu\nu} n_\nu = u^\mu (b^\nu n_\nu) - b^\mu (u^\nu n_\nu) = B_n \, u^\mu + \frac{J}{\rho} \, b^\mu \equiv C^\mu. \tag{4.7}$$

The RH conditions are therefore $[\![J]\!] = 0$, $[\![-J h_t u^\mu + P n^\mu - B_n b^\mu]\!] = 0$ and $[\![C^\mu]\!] = 0$.

4.3 Contact and tangential discontinuities

Proposition 4.1. *Let $J = 0$.*

(a) If $B^x \neq 0$ (contact discontinuity), then $\boldsymbol{v}$, $\boldsymbol{B}$, P and p are continuous; only ρ (and with it ϵ, h) can jump.

(b) If $B^x = 0$ (tangential discontinuity), then v_x and P are continuous, and ρ, p, the tangential velocity and the tangential field can all jump.

Proof. With $J = 0$, (4.4) gives $B_n = W_s B^x / W$ and (4.7) gives $C^\mu = (W_s B^x/W) u^\mu = W_s B^x (1, \boldsymbol{v})$. (a) $[\![C^\mu]\!] = 0$ and $B^x \neq 0$ give $[\![\boldsymbol{v}]\!] = 0$, hence $[\![W]\!] = 0$ and $[\![B_n]\!] = 0$. The momentum condition reduces to $[\![P]\!] n^\mu = B_n [\![b^\mu]\!]$; contracting with n_μ gives $[\![P]\!] = B_n [\![B_n]\!] = 0$, and then $[\![b^\mu]\!] = 0$. Since $B^i = u^0 b^i - u^i b^0$ with u^μ continuous, $[\![\boldsymbol{B}]\!] = 0$; b^2 is continuous, so $p = P - b^2/2$ is too. (b) With $B^x = 0$, $B_n = 0$ and $C^\mu = 0$ identically; the momentum condition is $[\![P]\!] n^\mu = 0$. The definition of J gives $V_s = v_{x,a} = v_{x,b}$. Nothing else is constrained. □

Across a contact the solution is thus fixed by (ρ_a, ρ_b) alone. Across a tangential discontinuity the gas pressure may jump as long as the total pressure does not: magnetic pressure can balance gas pressure. This is the central wave of the three-wave fan for $B^x = 0$ (Section 7.2).

4.4 Three invariants of discontinuities with $J \neq 0$

Proposition 4.2 (Shock invariants)**.** *Across any discontinuity with $J \neq 0$,*

$$[\![H]\!] = 0, \qquad H \equiv \frac{B_n^2}{J^2} - \frac{b^2}{\rho^2}, \tag{4.8}$$

$$[\![h \, B_n]\!] = 0, \tag{4.9}$$

$$[\![P]\!] + J^2 \left[\!\!\left[\frac{h}{\rho} \right]\!\!\right] = 0, \tag{4.10}$$

where h is the specific enthalpy of the gas and P the total pressure.

Proof. C^μ in (4.7) is continuous, and so is every scalar built from it and from other continuous vectors. Using $u_\mu u^\mu = -1$, $u_\mu b^\mu = 0$, $u_\mu n^\mu = -J/\rho$, $b_\mu n^\mu = B_n$, $n_\mu n^\mu = 1$:

- $C_\mu C^\mu = J^2 b^2/\rho^2 - B_n^2 = -J^2 H$. Since J is continuous and nonzero, $[\![H]\!] = 0$.
- Contract C_μ with the continuous vector $T^{\mu\nu} n_\nu$ of (4.6): $C_\mu u^\mu = -B_n$, $C_\mu n^\mu = B_n J/\rho - B_n J/\rho = 0$, $C_\mu b^\mu = J b^2/\rho$, so $C_\mu T^{\mu\nu} n_\nu = J B_n (h_t - b^2/\rho) = J B_n h$. Hence $[\![h B_n]\!] = 0$.
- Contract (4.6) with n_μ: $n_\mu T^{\mu\nu} n_\nu = J^2 h_t/\rho + P - B_n^2$, which is continuous. Writing $h_t/\rho = h/\rho + b^2/\rho^2$, $[\![P]\!] + J^2 [\![h/\rho]\!] - J^2 [\![B_n^2/J^2 - b^2/\rho^2]\!] = 0$, and the last term vanishes by (4.8). □

Equation (4.10) is eq. (4.26) of Giacomazzo and Rezzolla (2006), who quote the invariance of H from Anile (1989); they use (4.9) in their B_t-method. Remarkably, (4.10) has the same form as in hydrodynamics, with the total pressure but the *gas* enthalpy. A second relation, the *Lichnerowicz adiabat*, follows from the remaining jump conditions; in our notation it reads (Giacomazzo and Rezzolla, 2006, eq. 4.27)

$$[\![h^2]\!] - \left(\!\left(\frac{h}{\rho}\right)\!\right)[\![P]\!] - H[\![b^2]\!] + 2\left[\!\left[\frac{b^2 h}{\rho}\right]\!\right] - 2J^2 H\left[\!\left[\frac{h}{\rho}\right]\!\right] = 0. \tag{4.11}$$

We have not reproduced its derivation, but all of (4.8)–(4.11) were checked to 40 digits on a fast shock constructed numerically from the full RH system (Chapter D). For $\boldsymbol{b} = 0$, $H = 0$ and (4.11) is the Taub adiabat of Section 4.6.1.

4.5 Rotational (Alfvén) discontinuities

Proposition 4.3 (Alfvén discontinuities)**.** *Suppose $J \neq 0$ and ρ, p and b^2 are continuous (hence h, w and P are). Then either the discontinuity is trivial ($[\![u^\mu]\!] = 0$) or*

$$J^2 = \frac{\rho^2 B_n^2}{w}. \tag{4.12}$$

Writing $J = s\rho B_n/\sqrt{w}$ with $s = \pm 1$, the four-vector

$$\sigma^\mu \equiv b^\mu + s\sqrt{w}\, u^\mu \tag{4.13}$$

is continuous across the front, and the front moves with the Alfvén speed of either state,

$$V_s = \frac{\sigma^x}{\sigma^0} = \frac{b^x + s\sqrt{w}\, u^x}{b^0 + s\sqrt{w}\, u^0} = \lambda_A^{(s)}. \tag{4.14}$$

Proof. From (4.9) with h continuous, B_n is continuous. The momentum and Maxwell conditions become $-Jh_t[\![u^\mu]\!] - B_n[\![b^\mu]\!] = 0$ and $B_n[\![u^\mu]\!] + (J/\rho)[\![b^\mu]\!] = 0$. Eliminating $[\![b^\mu]\!] = -(\rho B_n/J)[\![u^\mu]\!]$ gives $(J^2 h_t - \rho B_n^2)[\![u^\mu]\!] = 0$, i.e. (4.12) since $\rho h_t = w$. Then $\rho B_n/J = s\sqrt{w}$ and $[\![b^\mu + s\sqrt{w} u^\mu]\!] = 0$. Finally, $J = s\rho B_n/\sqrt{w}$ with (4.4) reads $\sqrt{w}\, W(V_s - v_x) = s(b^x - V_s b^0)$, whose solution is (4.14); it can be evaluated on either side because σ^μ is continuous. □

The continuity of σ^μ is the relativistic Walén relation. In the Newtonian limit $\sigma^\mu \to (s\sqrt{\rho},\ s\sqrt{\rho}\,\boldsymbol{v} + \boldsymbol{B})$, so $\boldsymbol{v} + s\boldsymbol{B}/\sqrt{\rho}$ and the normal velocity are continuous and the tangential field rotates rigidly. In SRMHD the consequences are different (Komissarov, 1997); in the words of later summaries (Mignone and Bodo, 2006; Mignone et al., 2009), the thermodynamic quantities stay continuous, but the tangential field components trace ellipses rather than circles and the normal velocity v_x jumps. The spatial vector $\boldsymbol{K} \equiv \boldsymbol{\sigma}/\sigma^0$ is continuous, and its x-component is the Alfvén speed; this is the invariant on which the HLLD solver of Section 9.3 is built.

Example 4.1 (A relativistic rotational discontinuity)**.** Chapter D constructs rotational discontinuities numerically: fix the state ahead, compute σ^μ, pick u^z behind, and solve $u_\mu \sigma^\mu = -s\sqrt{w}$ (which is $u_\mu b^\mu = 0$) together with $[\![B^x]\!] = 0$ for u^x, u^y behind. All seven RH conditions then hold to 10^{-40} with $V_s = \lambda_A^{(s)}$. For the ahead state $(\rho, P, \boldsymbol{v}, \boldsymbol{B}) = (1,\ 2,\ (0.2, 0.3, -0.1),\ (0.8, 0.7, 0.4))$, $\Gamma = 5/3$, one left-going example has v_x changing by 8.9×10^{-3} and $|B_t|$ by -2.3% across the front, while ρ, p and b^2 are unchanged.

Coplanar data. If the tangential fields and velocities of the two states lie in one plane containing the normal (as in every face of a two-dimensional flow), the only rotational discontinuities that keep the solution in that plane are the trivial one and a reversal of the tangential field. In the solver record (§2) this is the reason the Alfvén unknowns degenerate on 2D runs: a continuous angle collapses to a binary choice.

In the code: Alfvén discontinuities

`rmhd/alfven_solver.py` and the batched `rmhd/batched/alfven_b.py` hold ρ and P fixed and solve eqs. (4.15)–(4.17), (4.19)–(4.20) of Giacomazzo and Rezzolla (2006) for $(v_x, v_y, v_z, B_y, B_z)$ behind, at $V_s = \lambda_A$, by Newton's method. The system has more than one root, and roots that satisfy the reduced equations but violate conservation exist; candidate roots are therefore ranked by the worst normalised residual of the seven RH conditions (`rmhd/batched/jump.py`, scaled by the flux magnitudes, not by the jump, so that weak discontinuities are not penalised).

4.6 Shocks

4.6.1 The relativistic hydrodynamic Taub adiabat

For $\boldsymbol{b} = 0$ the momentum condition is $[\![-Jhu^\mu + p\,n^\mu]\!] = 0$, i.e. $J[\![hu^\mu]\!] = [\![p]\!]n^\mu$.

Proposition 4.4 (Taub adiabat)**.** *Across a relativistic hydrodynamic shock (Taub, 1948)*

$$[\![h^2]\!] = \Big(\frac{h_a}{\rho_a} + \frac{h_b}{\rho_b}\Big)[\![p]\!], \qquad J^2 = -\frac{[\![p]\!]}{[\![h/\rho]\!]}. \tag{4.15}$$

Proof. Contract $J(h_a u_a^\mu - h_b u_b^\mu) = [\![p]\!]n^\mu$ with $h_a u_{a\mu} + h_b u_{b\mu}$. The left side is $J(-h_a^2 + h_b^2)$, the cross terms cancelling. The right side is $[\![p]\!](h_a u_a \cdot n + h_b u_b \cdot n) = -[\![p]\!]J(h_a/\rho_a + h_b/\rho_b)$. Divide by $-J$. The second relation is the n_μ-contraction, $[\![J^2 h/\rho + p]\!] = 0$, i.e. (4.10) with $\boldsymbol{b} = 0$. □

For a given ahead state and post-shock pressure, (4.15) with the EOS determines h_b and ρ_b; for the ideal gas it is a quadratic for h_b (Chapter 6). In the Newtonian limit $h \to 1+\epsilon+p/\rho$ with $\epsilon, p/\rho \ll 1$ and (4.15) reduces to the Hugoniot relation $[\![\epsilon]\!] = -\frac{1}{2}(p_a + p_b)[\![1/\rho]\!]$ (Exercise 4.2). Thorne (1973) discusses the properties of the Taub adiabat.

4.6.2 Magnetosonic shocks

Fast and slow shocks have $J \neq 0$, $[\![\rho]\!] \neq 0$, and all components may jump. Given the ahead state and the post-shock *total* pressure P_b, the jump conditions can be solved in closed form for the post-shock state as a function of V_s (Giacomazzo and Rezzolla, 2006, eqs. 4.13–4.25 and Appendix A); V_s is then the root of (4.10), which must be found numerically. This is the "p-method". The admissible windows for the root are set by the characteristic speeds of the ahead state: a right-going fast shock has $V_s \in (\lambda_A^+, 1)$, a right-going slow shock $V_s \in (v_x, \lambda_A^+)$, and symmetrically on the left.

In the code: shock speeds

`rmhd/shock_speed.py` evaluates the residual $f(V_s) = [\![P]\!] + J^2[\![h/\rho]\!]$, building the post-shock state with `rmhd/postshock.py` (`postshock_pressure`, a translation of the original Fortran), and searches the windows $(-1, \lambda_A^-)$, $(\lambda_A^+, 1)$, (λ_A^-, v_x), (v_x, λ_A^+) for the left fast, right fast, left slow and right slow shock. Because f can have several roots and NaN bands (where the post-shock state is unphysical), the window is scanned on fixed nodes, refined

near its edges, and the bracketed root is polished by a NaN-tolerant bisection. The solver record (§5a) documents a case where a pair of roots closer than one scan step was missed and the fix (a pair-splitting scan), and that on 119 of 120 such pairs exactly one root was an evolutionary fast shock.

Example 4.2 (A relativistic fast shock)**.** Take the ahead state with $\rho = 1$, $P = 2$ (gas pressure $p = 0.842$), $\boldsymbol{v} = (0.1, 0.2, 0.05)$, $\boldsymbol{B} = (1.2, 0.9, 0.3)$ and $\Gamma = 5/3$, and impose $P_b = 3$ behind a right-going shock. Solving the seven RH conditions for the state behind and V_s (Chapter D) gives $V_s = 0.86778$ and

$$(\rho, v_x, v_y, v_z, B^y, B^z)_b = (1.2966,\ 0.2911,\ 0.1262,\ 0.0291,\ 1.3518,\ 0.4430).$$

The right-going fast speeds are $\lambda_{\mathrm{f}}^{+} = 0.8275$ ahead and 0.8943 behind, and the right-going Alfvén speed behind is 0.6128, so $\lambda_{\mathrm{f}}^{+}(\boldsymbol{U}_a) < V_s < \lambda_{\mathrm{f}}^{+}(\boldsymbol{U}_b)$ and $V_s > \lambda_{\mathrm{A}}^{+}(\boldsymbol{U}_b)$: an evolutionary $1 \to 2$ fast shock (Section 4.7). The tangential field grows from $|B_t| = 0.949$ to 1.422, and the invariants of Proposition 4.2 and the Lichnerowicz adiabat hold to 10^{-39}.

What a shock does to the tangential field. In Newtonian MHD a fast shock increases $|B_t|$ and a slow shock decreases it; the limiting cases are the *switch-on* shock (fast, with $B_t = 0$ ahead) and the *switch-off* shock (slow, with $B_t = 0$ behind). Komissarov (2003) showed that relativistic switch-on and switch-off shocks exist as limits of fast and slow shocks, as in the Newtonian case.[1] A lab-frame rule "slow shocks weaken $|B_t|$" is, however, not reliable in SRMHD when the flow has a tangential velocity: the solver record (§5a) documents a slow shock that *strengthens* the lab-frame tangential field (tangential velocity 0.52) and a fast shock that weakens it. A solver that chooses between shock and rarefaction by the sign of the change of $|B_t|$ inherits this problem (Section 8.5).

4.7 Admissibility

The RH conditions admit discontinuities that are not physical, for instance rarefaction shocks. Two classical criteria select the admissible ones.

Lax conditions. A discontinuity is a Lax k-shock if the characteristics of family k run into it from both sides, while the characteristics of the other families cross it (Lax, 1957). For a right-going shock of family k this means $\lambda_k(\boldsymbol{U}_a) < V_s < \lambda_k(\boldsymbol{U}_b)$, $\lambda_{k-1}(\boldsymbol{U}_b) < V_s$ and $V_s < \lambda_{k+1}(\boldsymbol{U}_a)$.

Evolutionary condition. Perturb a discontinuity slightly. The linearised jump conditions are equations for the amplitudes of the *outgoing* small-amplitude waves (those carried away from the front on either side) and for the perturbation of the front speed; the incoming waves are given. The problem has a unique solution only if the number of unknowns equals the number of equations (Jeffrey and Taniuti, 1964; Falle and Komissarov, 2001). For a shock whose two states are coplanar—always the case in Newtonian MHD, where shocks do not rotate the tangential field—the reflection symmetry $z \to -z$ splits the linearised problem into two independent parts: the out-of-plane (Alfvénic) part, two equations (z-momentum and B^z) with no front-speed perturbation, which needs exactly *two* outgoing Alfvén waves; and the in-plane part, five equations for the outgoing magnetosonic and entropy waves plus the

[1] **Not independently verified:** the content of this statement was checked against the abstract of the preprint arXiv:astro-ph/0206076 by the same author on limit shocks of relativistic MHD; the text of the journal paper Komissarov (2003) was not accessed.

transition (ahead→behind)	name	outgoing: Alfvén + in-plane	ideal-MHD status
$1 \to 2$	fast shock	$2 + 4$	evolutionary
$3 \to 4$	slow shock	$2 + 4$	evolutionary
$1 \to 3$	intermediate	$1 + 4$	non-evolutionary
$1 \to 4$	intermediate	$1 + 3$	non-evolutionary
$2 \to 3$	intermediate	$1 + 5$	non-evolutionary
$2 \to 4$	intermediate	$1 + 4$	non-evolutionary
required		$2 + 4$	

Table 4.1: Counting of outgoing characteristics for a right-going coplanar shock. The rotational discontinuity ($V_s = \lambda_A$, linearly degenerate field) is not in this table: it is neither compressive nor of Lax type.

front speed, which needs exactly *four*. Classify each side of a right-going front by where V_s lies among that state's right-going speeds:

$$1\text{: } V_s > \lambda_{\mathrm{f}}^{+}, \quad 2\text{: } \lambda_{\mathrm{A}}^{+} < V_s < \lambda_{\mathrm{f}}^{+}, \quad 3\text{: } \lambda_{\mathrm{s}}^{+} < V_s < \lambda_{\mathrm{A}}^{+}, \quad 4\text{: } \lambda_{\mathrm{e}} < V_s < \lambda_{\mathrm{s}}^{+}. \tag{4.16}$$

An outgoing characteristic is one faster than V_s on the ahead side or slower than V_s on the behind side. Counting them (Table 4.1) shows that only the transitions $1 \to 2$ (fast shock) and $3 \to 4$ (slow shock), ahead state first, have the right numbers. The *intermediate* shocks $1 \to 3$, $1 \to 4$, $2 \to 3$ and $2 \to 4$ all have one outgoing Alfvén wave instead of two and are non-evolutionary in ideal MHD (for $2 \to 3$ the total is six, but split as $1 + 5$). Switch-on and switch-off shocks are limiting cases of fast and slow shocks.

The status of intermediate shocks. Whether intermediate shocks are physical has been debated for decades. Wu (1987, 1988, 1990) showed in dissipative simulations that they form from smooth waves and possess stable structures. Falle and Komissarov (2001) argued from the relation between evolutionary conditions and dissipative shock structure that non-evolutionary shocks are inadmissible, explained the intermediate shocks seen in planar MHD simulations, and suggested modifications of numerical schemes to avoid them in symmetric problems. Inoue and Inutsuka (2007) found that in dissipative MHD all shock types, including intermediate shocks, are evolutionary. Exact solvers take different positions: Giacomazzo and Rezzolla (2006) exclude compound waves and intermediate shocks by construction, whereas the Newtonian solver of Takahashi and Yamada (2014) includes all types of intermediate shocks and switch-on/off waves and can find all solutions of a given problem. We return to the consequences in Chapter 8.

In the code: which shock roots are accepted

The solver record (§5a) traced 120 fast waves on which the standard scan had found no shock: each had a pair of roots, and on 119 of them exactly one root was evolutionary (two-sided Lax on the fast family *and* super-Alfvénic behind, i.e. $1 \to 2$), the other leaving the flow sub-Alfvénic behind (an intermediate shock). The Lax condition on the fast family alone, which is what the scan checks, admits both. The pair-splitting scan (`RMHD_FAST_EDGE_SCAN`, on by default) runs on the lanes the standard scan gave up on; it splits the pair where the shock speed equals the Alfvén speed behind and takes the first root outward from there. An evolutionary selection of slow-shock roots exists but is parked (`RMHD_EVOLUTIONARY`). Every accepted answer must in addition pass the full RH residual and the wave-order gate (Section 7.4).

Exercises

Exercise 4.1. Verify that $T^{\mu\nu}n_\nu = W_s(T^{\mu x} - V_s T^{\mu 0})$ and hence that $[\![T^{\mu\nu}]\!]n_\nu = 0$ is equivalent to the momentum and energy rows of (4.1). Derive (4.5).

Exercise 4.2. Show that (4.15) reduces to the Newtonian Hugoniot relation $\epsilon_b - \epsilon_a = \frac{1}{2}(p_a + p_b)(1/\rho_a - 1/\rho_b)$ when ϵ and p/ρ are small.

Solution. Write $h = 1 + \epsilon + p/\rho$; to first order in small quantities $h_a^2 - h_b^2 \approx 2(\epsilon_a - \epsilon_b) + 2(p_a/\rho_a - p_b/\rho_b)$ and $(h_a/\rho_a + h_b/\rho_b)(p_a - p_b) \approx (1/\rho_a + 1/\rho_b)(p_a - p_b)$. Rearranging gives the Hugoniot relation.

Exercise 4.3. Show that the Lichnerowicz adiabat (4.11) reduces to (4.15) for $\boldsymbol{b} = 0$, and that for an Alfvén discontinuity ($[\![\rho]\!] = [\![p]\!] = [\![b^2]\!] = 0$) it is satisfied identically.

Exercise 4.4. Starting from (4.13), show that $u_\mu \sigma^\mu = -s\sqrt{w}$ and $\sigma_\mu \sigma^\mu = b^2 - w = -\rho h$ on both sides of a rotational discontinuity, and explain why these are consistent with the continuity of σ^μ.

Solution. $u_\mu \sigma^\mu = u_\mu b^\mu + s\sqrt{w}\, u_\mu u^\mu = -s\sqrt{w}$, and $\sigma_\mu \sigma^\mu = b^2 + 2s\sqrt{w}\, u_\mu b^\mu + w\, u_\mu u^\mu = b^2 - w$. Both sides have the same w and b^2, as required.

Exercise 4.5. Reproduce Table 4.1. Why is the total count of outgoing characteristics not a sufficient test for coplanar shocks?

Exercise 4.6. In the Newtonian limit, show that (4.13) gives the Walén relation $[\![\boldsymbol{v} + s\boldsymbol{B}/\sqrt{\rho}]\!] = 0$ and $[\![|B_t|]\!] = 0$ for a rotational discontinuity.

Chapter 5

Rarefaction waves

5.1 Self-similar simple waves

A rarefaction is a continuous self-similar solution $\boldsymbol{U} = \boldsymbol{U}(\xi)$, $\xi = x/t$. Substituting into $\partial_t \boldsymbol{U} + \mathsf{A}\,\partial_x \boldsymbol{U} = 0$ with $\partial_t = -(\xi/t)\mathrm{d}/\mathrm{d}\xi$ and $\partial_x = t^{-1}\mathrm{d}/\mathrm{d}\xi$ gives

$$\big(\mathsf{A}(\boldsymbol{U}) - \xi\,\mathsf{I}\big)\,\frac{\mathrm{d}\boldsymbol{U}}{\mathrm{d}\xi} = 0. \tag{5.1}$$

A non-constant solution therefore needs $\xi = \lambda_k(\boldsymbol{U}(\xi))$ for some family k, with $\mathrm{d}\boldsymbol{U}/\mathrm{d}\xi$ parallel to the right eigenvector $\boldsymbol{r}_k$. Differentiating $\xi = \lambda_k(\boldsymbol{U}(\xi))$ gives $1 = \nabla_{\boldsymbol{U}}\lambda_k \cdot \mathrm{d}\boldsymbol{U}/\mathrm{d}\xi$, hence

$$\boxed{\frac{\mathrm{d}\boldsymbol{U}}{\mathrm{d}\xi} = \frac{\boldsymbol{r}_k(\boldsymbol{U})}{\nabla_{\boldsymbol{U}}\lambda_k \cdot \boldsymbol{r}_k(\boldsymbol{U})}} \tag{5.2}$$

This requires $\nabla_{\boldsymbol{U}}\lambda_k \cdot \boldsymbol{r}_k \neq 0$ along the wave: the k-th field must be genuinely nonlinear there. Linearly degenerate fields (entropy, Alfvén) have no rarefactions; their simple waves are discontinuities moving at the characteristic speed. A k-rarefaction joining a left state $\boldsymbol{U}_l$ to a right state $\boldsymbol{U}_r$ occupies $\lambda_k(\boldsymbol{U}_l) \le \xi \le \lambda_k(\boldsymbol{U}_r)$ and is admissible only if λ_k increases monotonically through the fan. The edge facing the undisturbed state is the *head*, the other the *tail*.

Along the integral curve any function $w(\boldsymbol{U})$ with $\nabla_{\boldsymbol{U}} w \cdot \boldsymbol{r}_k = 0$ is constant: a *k-Riemann invariant*. An m-component system has $m-1$ independent ones for each family. In practice it is easier to parametrise the curve by a physical variable, e.g. the pressure: $\mathrm{d}\boldsymbol{U}/\mathrm{d}p = \boldsymbol{r}_k / r_k^{(p)}$, where $r_k^{(p)}$ is the pressure component of $\boldsymbol{r}_k$ in primitive variables.

5.2 Relativistic hydrodynamics

For normal flow ($v_y = v_z = 0$) take $\boldsymbol{V} = (\rho, v, p)$ with $v \equiv v_x$. The eigenvalues of $\tilde{\mathsf{A}}$ are

$$\lambda_{\mathrm{e}} = v, \qquad \lambda_{\pm} = \frac{v \pm c_s}{1 \pm v c_s}, \tag{5.3}$$

the relativistic composition of the flow speed with $\pm c_s$, and along the $\pm$ acoustic eigenvectors

$$\frac{\mathrm{d}v}{\mathrm{d}p} = \pm\frac{1}{W^2 \rho h c_s}, \qquad \mathrm{d}p = h c_s^2\,\mathrm{d}\rho \tag{5.4}$$

(both checked from the Jacobian in Chapter D). With the rapidity $\vartheta = \operatorname{artanh} v$, $\mathrm{d}\vartheta = W^2 \mathrm{d}v$, so (5.4) becomes $\mathrm{d}\vartheta = \pm \mathrm{d}p/(\rho h c_s) = \pm c_s\,\mathrm{d}\rho/\rho$. The Riemann invariants are therefore

$$\mathcal{J}_{\pm} = \vartheta \pm \int \frac{c_s}{\rho}\,\mathrm{d}\rho, \tag{5.5}$$

with $\mathcal{J}_+$ constant across a left-going (λ_-) rarefaction and $\mathcal{J}_-$ constant across a right-going (λ_+) one.

Ideal gas. On the isentrope $p = K\rho^\Gamma$ one finds $c_s\,\mathrm{d}\rho/\rho = 2\,\mathrm{d}c_s/(\Gamma - 1 - c_s^2)$ (checked symbolically), so

$$\int \frac{c_s}{\rho}\mathrm{d}\rho = \mathcal{G}(c_s) \equiv \frac{1}{\sqrt{\Gamma-1}} \ln \frac{\sqrt{\Gamma-1}+c_s}{\sqrt{\Gamma-1}-c_s}, \tag{5.6}$$

and across a rarefaction facing the state a (sign $s = -1$ for a left-going wave, $+1$ for a right-going one)

$$\operatorname{artanh} v_b = \operatorname{artanh} v_a + s\,[\mathcal{G}(c_{s,b}) - \mathcal{G}(c_{s,a})], \qquad \rho_b = \rho_a \Big(\frac{p_b}{p_a}\Big)^{1/\Gamma}. \tag{5.7}$$

Inside the fan the state at ξ follows from $\xi = \lambda_s(v, c_s)$ together with (5.7).

Tangential velocities. With $v_t \neq 0$ the tangential momentum equation gives one more invariant. For the self-similar flow, $\mathrm{d}(S_t/D) = 0$ across the wave, i.e.

$$A \equiv hW v_t = \text{const}, \qquad \text{hence} \qquad v_t^2 = \frac{A^2(1-v_x^2)}{h^2 + A^2}, \tag{5.8}$$

and the acoustic ODE becomes (Pons et al., 2000)

$$\frac{\mathrm{d}v_x}{\mathrm{d}p} = s\,\frac{(1-v_x^2)\sqrt{h^2 + A^2(1-c_s^2)}}{(h^2+A^2)\,\rho\,c_s}, \tag{5.9}$$

which reduces to (5.4) for $A = 0$. We checked (5.9) and the invariance of A against the eigenvectors of the four-variable Jacobian (Chapter D). The tangential velocity enters through $h^2 + A^2$: unlike in Newtonian hydrodynamics, the normal and tangential motions are coupled through the Lorentz factor.

5.3 Relativistic MHD

For SRMHD the self-similar equations (5.1) are seven ODEs for $(\rho, P, v_x, v_y, v_z, B^y, B^z)$, written out by Giacomazzo and Rezzolla (2006, eqs. 4.43–4.49). Non-trivial solutions need ξ to be a root of the magnetosonic quartic (3.15) (their eq. 4.50): fans exist only in the fast and slow families. There is no closed form for the integral curves, and Giacomazzo and Rezzolla (2006) integrate them numerically with the total pressure as the independent variable (six ODEs; their Appendix B). Two structural facts are worth deriving.

The Alfvén singularity. The right-hand sides of Appendix B of Giacomazzo and Rezzolla (2006) contain the factor

$$\mathcal{R} = \frac{1}{\rho h W^4 (\eta^2 - w)\,(\lambda_A^{(+)} - \xi)(\lambda_A^{(-)} - \xi)} \tag{5.10}$$

and, in the equations for $\mathrm{d}B^y/\mathrm{d}P$ and $\mathrm{d}B^z/\mathrm{d}P$, the denominator

$$(B^x)^2 + 2B^x \eta W^2 (v_x - \xi) + W^4(\eta^2 - w)(v_x - \xi)^2 = W^4(\eta^2 - w)\,(\lambda_A^{(+)} - \xi)(\lambda_A^{(-)} - \xi). \tag{5.11}$$

The factorisation follows by viewing the left side as a quadratic in $W^2(v_x - \xi)$ with roots $-B^x/(\eta \pm \sqrt{w})$, i.e. $\xi = \lambda_A^{(\pm)}$ by (3.14); it was checked symbolically. So the fan equations are singular exactly where the fan speed ξ (a fast or slow speed) equals an Alfvén speed: at a Type II degeneracy. Giacomazzo and Rezzolla (2006) note that such points cannot be integrated without a regularisation, which their code does not include. Since $\lambda_s \le \lambda_A \le \lambda_f$, a slow fan approaches this singularity when the field becomes nearly normal to the front.

Other parametrisations. Between the two slow waves GR06 use the tangential field as the unknown. Their slow fans are integrated with $|B_t|$ as the independent variable at fixed angle $\psi = \arctan(B^z/B^y)$, and, when the field rotates inside the slow fan (as in Balsara's test 5), with ψ itself (Giacomazzo and Rezzolla, 2006, §5.2). Note that in SRMHD ψ need not be constant across fast and slow waves, unlike in Newtonian MHD.

End points and admissibility. A fan is integrated from the head (the state it faces, at $\xi = \lambda_k(\boldsymbol{U}_a)$) until the independent variable reaches its target; the tail speed is λ_k of the end state. The fan is admissible only if λ_k is monotone along it. If it is not—because the field is not genuinely nonlinear along the path—the wave must be replaced by a compound structure (Chapter 8).

In the code: fans

`rmhd/rarefaction.py` integrates the fan ODEs with fixed-step fourth-order Runge–Kutta, with three independent variables: the total pressure (`'pressure'`, 800 steps), $|B_t|$ at fixed ψ (`'normB'`, 500 steps) and ψ (`'psi'`, 1000 steps). The right-hand sides contain the factors $(\lambda_A^{(+)} - \xi)(\lambda_A^{(-)} - \xi)$ discussed above. The batched versions are in `rmhd/batched/rarefaction_b.py` and `fan_njit.py`.
Which variable to use matters. The solver record (§5a) measured, along the slow family at the input states, $\mathrm{d}\ln P/\mathrm{d}\ln|B_t| \approx -0.005$ (median) on interfaces the five-wave solver answers, and -0.54 to -0.69 on unsolved interfaces where the field is nearly normal to the face: there $|B_t|$ hardly moves along the slow family, so a slow wave steered by $|B_t|$ is steered by a coordinate that does not change. The four-unknown solver (Section 7.4) therefore chooses, for each slow wave, between B_t and P from the tangent of the slow family.

5.4 The state on the ray $\xi = 0$

Godunov's flux needs the state at $\xi = 0$. If $\xi = 0$ falls inside a fan, the integration is continued from the head only until $\lambda_k(\boldsymbol{U}) = 0$. In the code this is `state_at_xi` in `rmhd/batched/ray_b.py`, which also handles the case of a constant zone.

Exercises

Exercise 5.1. Show that (5.2) implies $\mathrm{d}\lambda_k/\mathrm{d}\xi = 1$ along the fan, and that a rarefaction across which λ_k would have to decrease is not a solution.

Exercise 5.2. Derive (5.4) directly from the self-similar form of the mass, momentum and energy equations of relativistic hydrodynamics (normal flow). *Hint:* the mass equation gives $(v-\xi)\rho' + \rho\, W^2(1 - v\xi)\, v' = 0$.

Exercise 5.3. Gas at rest with sound speed c_a fills $x > 0$; $x < 0$ is vacuum. The gas expands to the left through a right-going rarefaction. Show that the gas–vacuum interface moves with $v_{\mathrm{vac}} = -\tanh \mathcal{G}(c_a)$. Evaluate it for $\Gamma = 5/3$, $p/\rho = 1$, and compare with the Newtonian escape speed $2c_s/(\Gamma - 1)$ evaluated with the same c_s.

Solution. At the vacuum end $\rho \to 0$, so $c_s \to 0$ and $\mathcal{G} \to 0$; (5.7) gives $|\operatorname{artanh} v_{\mathrm{vac}}| = \mathcal{G}(c_a)$. For $\Gamma = 5/3$, $p/\rho = 1$: $h = 3.5$, $c_a^2 = 5/10.5$, $c_a \approx 0.690$, $\sqrt{\Gamma - 1} \approx 0.816$, so $\mathcal{G} \approx 1.225 \ln(1.506/0.126) \approx 3.04$ and $|v_{\mathrm{vac}}| \approx 0.995$. The Newtonian formula would give 2.07, faster than light.

Exercise 5.4. Show that $A = hWv_t$ is continuous across a relativistic hydrodynamic shock as well. *Hint:* use (4.5) on the tangential momentum.

Exercise 5.5. Prove the factorisation (5.11).

Chapter 6

The relativistic hydrodynamic Riemann problem

Before the seven-wave problem we solve the three-wave one completely. The method—express the state behind each outer wave as a function of the pressure behind it, then match at the contact—is exactly the one generalised to SRMHD in Chapter 7. It is due to Martí and Müller (1994) for normal flow and to Pons et al. (2000) for arbitrary tangential velocities; Rezzolla and Zanotti (2001); Rezzolla et al. (2003) added an a priori determination of the wave pattern.

6.1 Structure of the solution

The solution consists of a left-going wave (shock or rarefaction), a contact discontinuity, and a right-going wave:

$$L \longrightarrow \mathcal{W}_{\leftarrow} \longrightarrow L_* \longrightarrow \mathcal{C} \longrightarrow R_* \longrightarrow \mathcal{W}_{\rightarrow} \longrightarrow R. \tag{6.1}$$

Across the contact p and v_x are continuous (Proposition 4.1 with $\boldsymbol{b} = 0$), so the unknowns are p_* and v_*; ρ and the tangential velocity jump. Each outer wave is a shock if the pressure rises across it (p_* larger than the pressure of the state it faces) and a rarefaction otherwise.

6.2 The shock branch

Let a be the state the shock faces and p_b the pressure behind it. Using the mass flux $j \equiv J = \rho_a W_a W_s (V_s - v_{x,a})$ of (4.4) (positive for a right-going shock):

(i) *Post-shock enthalpy.* The Taub adiabat (4.15) with the ideal-gas relation $\rho_b = \Gamma p_b/[(\Gamma - 1)(h_b - 1)]$ is a quadratic for h_b,

$$\Big[1 + \frac{(\Gamma-1)(p_a - p_b)}{\Gamma p_b}\Big] h_b^2 - \frac{(\Gamma-1)(p_a - p_b)}{\Gamma p_b}\, h_b + \frac{h_a(p_a - p_b)}{\rho_a} - h_a^2 = 0, \tag{6.2}$$

whose positive root is the physical one.

(ii) *Mass flux.* $j^2 = (p_b - p_a)/(h_a/\rho_a - h_b/\rho_b)$, from (4.15).

(iii) *Shock speed.* Solving $j = \rho_a W_a W_s (V_s - v_{x,a})$ for V_s,

$$V_s = \frac{\rho_a^2 W_a^2 v_{x,a} + j\sqrt{j^2 + \rho_a^2 W_a^2 (1 - v_{x,a}^2)}}{\rho_a^2 W_a^2 + j^2}, \tag{6.3}$$

with $j > 0$ for a right-going and $j < 0$ for a left-going shock.

(iv) *Post-shock normal velocity.* The momentum condition with (4.5) gives $h_b W_b v_{x,b} = h_a W_a v_{x,a} + W_s(p_b - p_a)/j$, and the energy and mass conditions give $h_b W_b = h_a W_a + (p_b - p_a)[W_s v_{x,a}/j + 1/(\rho_a W_a)]$. Dividing,

$$v_{x,b} = \frac{h_a W_a v_{x,a} + W_s (p_b - p_a)/j}{h_a W_a + (p_b - p_a)[W_s v_{x,a}/j + 1/(\rho_a W_a)]}. \tag{6.4}$$

These are the relations of Pons et al. (2000) (see also eq. 4.13 of Romero et al. (2005), who write them for the magnetised case of Section 7.2.1). They hold with tangential velocities too, if W_a includes $v_{t,a}$; the tangential velocity behind follows from the invariance of $A = hWv_t$ (Exercise 6.3) and (5.8). A shock built from (i)–(iv) satisfies all jump conditions to 38 digits (Chapter D).

Derivation of (iv). With $S/D = hWv_x$ and $E/D = hW - p/D$, the momentum and energy jumps read $(j/W_s)[\![hWv_x]\!] = [\![p]\!]$ and $(j/W_s)[\![hW - p/D]\!] = [\![pv_x]\!]$, and mass conservation gives $1/D_b = 1/D_a + (W_s/j)(v_{x,a} - v_{x,b})$. Substituting the last relation into the energy jump, the terms in $v_{x,b}$ cancel and $h_b W_b$ follows; the momentum jump gives $h_b W_b v_{x,b}$.

6.3 The rarefaction branch

Across a rarefaction facing a, (5.7) gives $v_{x,b}$ in closed form for normal flow. With tangential velocities one integrates (5.9) from p_a to p_b along the isentrope.

6.4 The equation for the central pressure

Define, for the left and right waves,

$$v_{L*}(p) = \begin{cases} \mathcal{S}_L(p), & p > p_L \\ \mathcal{R}_L(p), & p \le p_L \end{cases}, \qquad v_{R*}(p) = \begin{cases} \mathcal{S}_R(p), & p > p_R \\ \mathcal{R}_R(p), & p \le p_R \end{cases}, \tag{6.5}$$

where $\mathcal{S}$ and $\mathcal{R}$ denote the shock and rarefaction branches of Section 6.2 and (5.7). The solution is the root of

$$\boxed{v_{L*}(p_*) = v_{R*}(p_*)} \tag{6.6}$$

v_{L*} decreases and v_{R*} increases monotonically with p, so for normal flow (6.6) has exactly one root, which bisection or a safeguarded Newton iteration finds reliably (Martí and Müller, 1994). Figure 6.1 shows the two curves for a strong blast wave.

Example 6.1 (Blast wave)**.** For the data of Fig. 6.1 the left wave is a rarefaction and the right wave a shock. Solving (6.6) in 30-digit arithmetic with the formulas above gives $p_* = 18.5970787$, $v_* = 0.96040961$, $\rho_{L*} = 0.0915518$, $\rho_{R*} = 10.41558$ and a shock speed $V_s = 0.98680$. The exact solver of the reference implementation (built-in case 3 of `rmhd/initial_data.py`, run with $B = 0$) returns $p_* = 18.5970777$ and $v_* = 0.96040961$; the relative difference in p_*, 5×10^{-8}, is consistent with its fixed-step integration of the fan and its 10^{-8} tolerance. The thin dense shell behind the shock ($\rho_{R*}/\rho_R \approx 10.4$, more than the Newtonian limit $(\Gamma+1)/(\Gamma-1) = 4$) is the hallmark of relativistic blast waves.

6.5 Wave patterns and the relative velocity

Which of the four patterns (shock–shock, shock–rarefaction, rarefaction–shock, rarefaction–rarefaction) occurs is decided by where p_* lies relative to p_L and p_R, and hence by the velocity

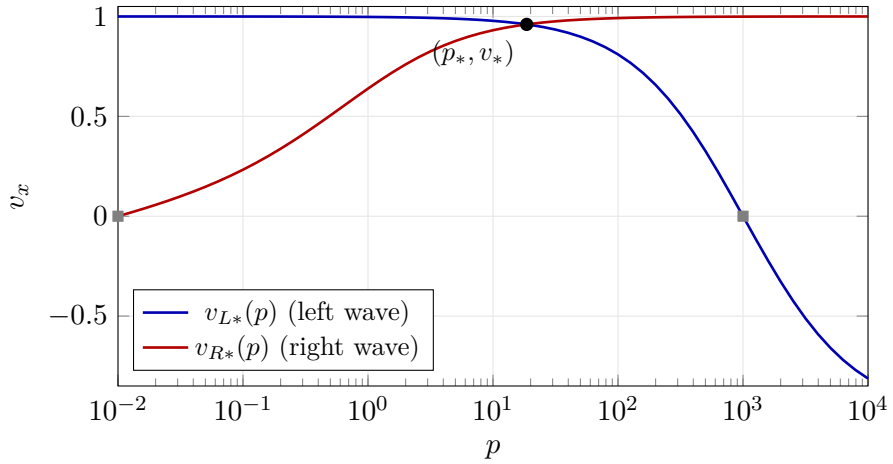


Figure 6.1: The functions $v_{L*}(p)$ and $v_{R*}(p)$ for the blast wave $(\rho, p, v_x)_L = (1, 1000, 0)$, $(\rho, p, v_x)_R = (1, 0.01, 0)$, $\Gamma = 5/3$. Each curve is a rarefaction branch below the pressure of its state (grey squares) and a shock branch above it. They cross at $p_* = 18.597$, $v_* = 0.96041$.

jump between the initial states. Because v_{L*} and v_{R*} are monotone, the limiting cases are obtained by evaluating them at $p = p_L$ and $p = p_R$. Rezzolla and Zanotti (2001) showed that the pattern is determined by the relativistic relative velocity of the initial states compared with such limiting values, which removes the initial guesswork, and Rezzolla et al. (2003) extended this to flows with tangential velocities.

6.6 Tangential velocities

With $v_t \neq 0$ the problem is still a two-unknown problem in (p_*, v_*), but the tangential velocity couples to the normal one through W: $A = hWv_t$ is invariant across shocks and rarefactions (not across the contact), the shock relations (6.3)–(6.4) use the full W_a, and the rarefaction branch uses (5.9). Large tangential velocities can change the wave pattern for fixed thermodynamic states, and the pattern changes continuously as v_t varies (Pons et al., 2000; Rezzolla et al., 2003). The tangential velocity changes its magnitude across a wave but, since $[\![S_y/D]\!] = [\![S_z/D]\!] = 0$, not its direction (Giacomazzo and Rezzolla, 2006, §5.2).

Exercises

Exercise 6.1. Derive (6.2) and show that its positive root satisfies $h_b > h_a$ when $p_b > p_a$.

Exercise 6.2. Derive (6.3). Show that $|V_s| \to 1$ as $|j| \to \infty$ and that $V_s \to v_{x,a}$ as $j \to 0$.

Exercise 6.3. Using (4.5), show that across a hydrodynamic shock $[\![hWv_t]\!] = 0$, and that the tangential velocity behind is $v_t^2 = A^2(1 - v_x^2)/(h^2 + A^2)$.

Solution. The tangential momentum jump is $V_s[\![S_t]\!] = [\![S_t v_x]\!]$, i.e. $[\![S_t(V_s - v_x)]\!] = 0 = (j/W_s)[\![S_t/D]\!]$ and $S_t/D = hWv_t$. Then $A^2 = h^2W^2v_t^2$ with $W^{-2} = 1 - v_x^2 - v_t^2$ gives the formula.

Exercise 6.4 (Programming)**.** Implement the solver of this chapter (normal flow) and reproduce Example 6.1. Then give the right state a tangential velocity $v_t = 0.9$ and observe how p_* and the pattern change.

Exercise 6.5. Show that for $p_L = p_R$ and $v_{x,L} > v_{x,R}$ the solution has two shocks, and for $v_{x,L} < v_{x,R}$ two rarefactions. What happens if $v_{x,R} - v_{x,L}$ is large enough that $p_* \to 0$?

Chapter 7

The SRMHD Riemann problem

7.1 Seven waves, eight states

For $B^x \neq 0$ and generic data the solution of the SRMHD Riemann problem consists of seven waves: a fast wave, an Alfvén discontinuity and a slow wave moving left, a contact discontinuity, and a slow wave, an Alfvén discontinuity and a fast wave moving right (Fig. 7.1). They separate eight constant states, $R_1 = L$, $R_2, \dots, R_7$, $R_8 = R$, in the numbering of Giacomazzo and Rezzolla (2006). Fast and slow waves are shocks or rarefactions; the Alfvén waves and the contact are discontinuities.

Counting unknowns. Each wave belongs to a one-parameter family of states reachable from the state ahead of it (a wave curve). Fixing one parameter per non-contact wave—six parameters—fixes $R_2, \dots, R_4$ from the left and $R_7, \dots, R_5$ from the right. The contact then imposes six conditions (Proposition 4.1): continuity of v_x, v_y, v_z, B^y, B^z and P. The exact Riemann problem is therefore a system of six nonlinear equations in six unknowns. Every exact method is a choice of these six parameters, and the difficulty lies in that choice: it decides how well conditioned the system is and where it is even defined.

7.2 Special cases

7.2.1 Purely tangential field orthogonal to the flow

Romero et al. (2005) solved the case $B^x = 0$ with $\boldsymbol{v} = (v_x, 0, v_z)$ and $\boldsymbol{B} = (0, B^y, 0)$, so that $\boldsymbol{v} \cdot \boldsymbol{B} = 0$. Then $b^0 = 0$ and $b^\mu = (0, 0, b, 0)$ with $b = B^y/W$, and (2.9) becomes

$$T^{\mu\nu} = \rho\hat{h}\, u^\mu u^\nu + \hat{p}\, g^{\mu\nu} - b^\mu b^\nu, \qquad \hat{h} = h + \frac{b^2}{\rho}, \qquad \hat{p} = p + \frac{b^2}{2}. \tag{7.1}$$

The last term has only a yy component, and $\partial_\mu(b^\mu b^\nu) = \partial_y(b\, b^\nu) = 0$ in a flow that depends on (t, x) only. The energy–momentum equations are therefore those of relativistic hydrodynamics with pressure $\hat{p}$ and enthalpy $\hat{h}$. The induction equation is $\partial_t B^y + \partial_x(v_x B^y) = 0$, and with mass conservation it says that $B^y/D = b/\rho$ is carried with the fluid; by (4.5) it is also continuous across shocks. So on each side of the contact $\hat{B} \equiv b/\rho$ is a constant, and the problem is relativistic hydrodynamics with the effective equation of state (Romero et al., 2005, Appendix A)

$$\hat{p} = p + \tfrac{1}{2}\hat{B}^2\rho^2, \qquad \hat{\epsilon} = \epsilon + \tfrac{1}{2}\hat{B}^2\rho, \qquad \hat{c}_s^2 = \frac{hc_s^2 + \hat{B}^2\rho}{\hat{h}} = c_s^2 + v_A^2 - c_s^2 v_A^2, \quad v_A^2 = \frac{b^2}{\rho\hat{h}}. \tag{7.2}$$

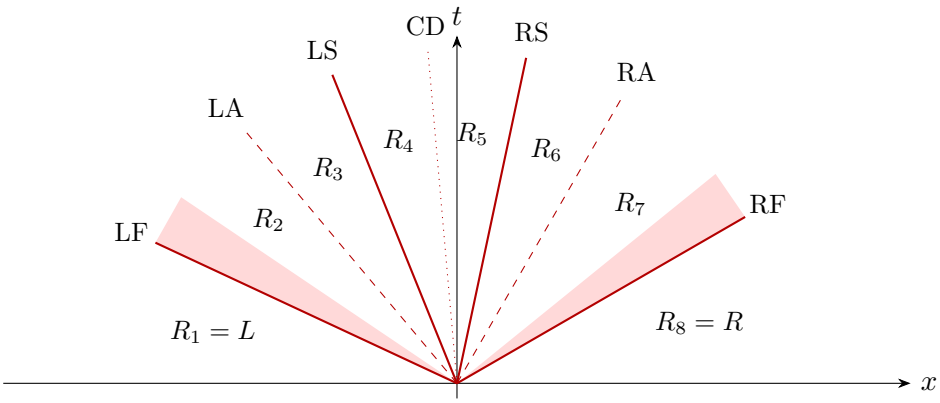


Figure 7.1: The seven-wave Riemann fan of SRMHD (fast: LF, RF; Alfvén: LA, RA; slow: LS, RS; contact: CD). The shaded wedges indicate that fast and slow waves may be rarefaction fans. For $B^x = 0$ the Alfvén and slow waves collapse onto the contact (three waves); for coplanar data the Alfvén waves are either absent or reverse the field.

The last identity (checked symbolically) shows that the effective sound speed is the fast speed ζ for propagation perpendicular to the field. Everything in Chapter 6 applies with hats: the three waves are fast shocks or rarefactions and a tangential discontinuity across which $\hat{p}$ and v_x are continuous. Magnetic pressure raises $\hat{p}$ in the initial states and can change the wave pattern for fixed gas states (Romero et al., 2005).

7.2.2 Normal field zero: three waves

For $B^x = 0$ but $\boldsymbol{v} \cdot \boldsymbol{B} \neq 0$ the reduction above fails, but the fan still has three waves (Type I degeneracy): two fast waves and a tangential discontinuity at which only v_x and P are continuous (Proposition 4.1). Giacomazzo and Rezzolla (2006) solve it like the hydrodynamic problem: the states behind the fast waves are functions of the total pressure behind them (the p-method of Section 4.6 and Section 5.3), and the single unknown P_* is the root of $v_{x,L*}(P_*) = v_{x,R*}(P_*)$.

7.2.3 Coplanar data: five waves

If the tangential fields and velocities of L and R lie in one plane containing the normal—as on every face of a two-dimensional flow—the solution stays in that plane. A rotational discontinuity can then only leave the field unchanged or reverse it (Section 4.5). The fan has five waves (fast, slow, contact, slow, fast), plus possibly a reversal at one or both Alfvén speeds, which is a discrete choice rather than a continuous parameter. With B_t a *signed* scalar, the unknowns are the two fast-wave strengths and the tangential field at the contact, and the contact conditions are $[\![v_x]\!] = [\![v_t]\!] = [\![P]\!] = 0$: three equations in three unknowns. This is the formulation of the planar solver below. Coplanar Riemann problems are also where the non-uniqueness of Chapter 8 lives.

7.3 The exact solution of Giacomazzo and Rezzolla

Giacomazzo and Rezzolla (2006) gave the first exact solution of the general problem. Its ingredients are:

- *the p-method* for the fast waves and the Alfvén discontinuities: the total pressure behind the fast wave is the unknown; the post-shock state follows from the jump conditions and the shock speed from (4.10); a fast rarefaction is integrated in P; across the Alfvén

	ρ	P	v_x	v_y	v_z	B^y	B^z
$R_1 = L$	1.000	78.50	0.0000	0.0000	0.0000	6.000	6.000
R_2	0.4300	23.21	0.6344	−0.09981	−0.09981	3.045	3.045
R_3	0.4300	23.21	0.6344	−0.09981	−0.09981	3.045	3.045
R_4	0.3830	22.84	0.6770	−0.05566	−0.05566	3.205	3.205
R_5	2.828	22.84	0.6770	−0.05566	−0.05566	3.205	3.205
R_6	1.582	20.72	0.4688	−0.2538	−0.2538	3.971	3.971
R_7	1.582	20.72	0.4688	−0.2538	−0.2538	3.971	3.971
$R_8 = R$	1.000	13.99	0.0000	0.0000	0.0000	0.7000	0.7000

Table 7.1: Exact solution of Balsara's test 2 ($B^x = 5$, $\Gamma = 5/3$), from Giacomazzo and Rezzolla (2006, Table 8); P is the total pressure.

discontinuity ρ and P are continuous and the remaining five jump conditions are solved by Newton's method at $V_s = \lambda_A$;

- *the B_t-method* for the slow waves: the tangential field between the slow waves is the unknown; across a slow shock the post-shock state follows from the jump conditions with (B^y, B^z) prescribed, the pressure from the invariance of hB_n (4.9), and the speed again from (4.10); a slow rarefaction is integrated in $|B_t|$ (or in ψ when the field rotates);
- *a 4×4 Newton–Raphson iteration* on the unknowns (P_L, B^y_c, B^z_c, P_R)—the total pressures behind the two fast waves and the tangential field at the contact—with the residuals $[\![v_x]\!], [\![v_y]\!], [\![v_z]\!], [\![P]\!]$ at the contact. Continuity of B^y, B^z is built in by the B_t-method. Using the total pressure as the unknown in all three pairs of regions instead would give three unknowns but five conditions at the contact; GR06 found that round-off alone then prevents the five conditions from being satisfied simultaneously.

They report that the residuals change very rapidly near the root, so the iteration converges only from a sufficiently close initial guess; they take it from an HLLE solution on about 800 cells. Their test suite—Komissarov's shock tubes, Balsara's five tests (Table 10.1) and a generic Alfvén test—is solved to residuals of about 10^{-10}. Two exceptions are instructive: Balsara's test 1 develops a compound wave in numerical solutions, which the method excludes by construction (Chapter 8); and in test 5 the left Alfvén and slow waves are so close that the Alfvén jump is resolved less accurately and the field rotates inside the slow rarefaction.

Example 7.1 (Balsara's test 2)**.** Table 7.1 reproduces the exact solution of Balsara's test 2 given by Giacomazzo and Rezzolla (2006, Table 8). Both initial states have $v_y = v_z$ (zero) and $B^y = B^z$, so the data are coplanar in the plane spanned by $\hat{\boldsymbol{x}}$ and $(\hat{\boldsymbol{y}} + \hat{\boldsymbol{z}})/\sqrt{2}$; accordingly $R_2 = R_3$ and $R_6 = R_7$: the Alfvén waves are silent and the fan has five waves. The left fast and slow waves are rarefactions, the right slow and fast waves are shocks, and across the contact only ρ jumps. The built-in initial guess for this case in `rmhd/initial_data.py` (case 9), $(P_{\mathrm{LF}}, B^y_c, B^z_c, P_{\mathrm{RF}}) = (23.214, 3.2052, 3.2052, 20.721)$, agrees with the table to the digits printed.

7.4 An exact solver used as a flux function

The reference implementation (`rmhd_final`) ports the method above to Python, batches it over many interfaces, and uses it as the flux of a two-dimensional constrained-transport code (`HLLD`). What follows is a summary of its own documentation (the guide, Parts 1, 3 and 4; the solver record, §§1–3, 5–5a); the numbers quoted are measurements recorded there, mostly on a 64^2 relativistic rotor.

State and frame. Every face is rotated so that its normal is x; a state is the vector $(\rho, P, v_x, v_y, v_z, B^y, B^z)$, with $B_n = B^x$ held separately. Faces are first classified (Section 3.5).

Seven-wave Newton. The production formulation has six unknowns and six residuals,

$$\boldsymbol{u}_6 = [\ln P_{\rm LF},\ \ln|B_{t,\rm CD}|,\ \psi_{\rm CD},\ \ln P_{\rm RF},\ \varphi_L,\ \varphi_R], \tag{7.3}$$

$$\boldsymbol{f}_6 = [[\![v_x]\!],\ [\![v_y]\!],\ [\![v_z]\!],\ [\![P]\!],\ s_L,\ s_R], \tag{7.4}$$

where $P_{\rm LF}, P_{\rm RF}$ are the total pressures behind the fast waves, $(|B_{t,\rm CD}|, \psi_{\rm CD})$ the tangential field at the contact, and $\varphi_{L,R}$ the rotation angles of the Alfvén waves. Each slow wave is told both components of the contact field but has one free parameter; the *slacks* $s_{L,R}$ measure how far it misses. The Jacobian is a finite difference whose six probes are stacked into the batch; the step comes from a singular value decomposition with small singular values truncated, followed by a line search. Converged lanes are frozen, and every lane's arithmetic is bit-for-bit what it would be alone (batch independence).

Seeds. Because the basin of convergence is narrow, as GR06 already found, the Newton is seeded by a small neural network that predicts $\boldsymbol{u}_6$ from the two states, written in a canonical form that removes the symmetries of the problem. Failed lanes are retried from other networks' seeds and from jittered seeds. The recorded experience is that convergence is threshold-shaped in the quality of the seed and that more *diverse* starts help where better single seeds do not (the guide, §1.9).

The five-wave planar solver. On a 2D flow every face is coplanar to about 10^{-10} (median of the cross product of the normalised tangential fields, 7.6×10^{-11} on the rotor). There the seven-wave formulation is ill-posed in the directions it does not need: $\psi_{\rm CD}$ is fixed by geometry, the slow waves are over-determined, and the Alfvén and slow speeds nearly coincide. The planar solver (`rmhd/batched/planar5_b.py`) rotates the frame so that the left tangential field points along $+y$ and solves

$$\boldsymbol{u}_3 = [\ln P_{\rm LF},\ B_{t,\rm CD},\ \ln P_{\rm RF}], \qquad \boldsymbol{f}_3 = [[\![v_x]\!],\ [\![v_t]\!],\ [\![P]\!]], \tag{7.5}$$

with $B_{t,\rm CD}$ signed, so that a field reversal through the contact is a sign change and a reversal at an Alfvén speed is a discrete branch (`flip`). Only a small part of the natural box of unknowns is constructible (about 6–8%), so the Jacobian falls back to backward differences where a forward probe leaves the feasible set.

The four-unknown planar solver. Where the field is nearly normal to the face and the Alfvén speed exceeds the sound speed, the fast waves carry the tangential field and the slow waves carry the pressure, so steering a slow wave by B_t fails (Chapter 5). The four-unknown solver (`rmhd/batched/planar4_b.py`) gives each wave its own strength,

$$\boldsymbol{u}_4 = [\ln P_{\rm LF},\ s_L,\ s_R,\ \ln P_{\rm RF}], \qquad \boldsymbol{f}_4 = [[\![v_x]\!],\ [\![v_t]\!],\ [\![B_t]\!]/\sqrt{2P},\ [\![P]\!]/P], \tag{7.6}$$

where s is the signed tangential field (kind B) or $\ln P$ (kind P) behind the slow wave, chosen per wave from the tangent of the slow family. It starts from slow waves of zero strength and carries its own certificate: every shock satisfies the full RH conditions to 10^{-8}, every fan is an integral of the fan ODE, the contact closes to 10^{-8}, the input is planar and the fan is ordered.

what happens to a face (all 8.0×10^6 face evaluations)		share
below the weak-jump gate (linearised flux)		85.9%
fan entirely on one side (upwind)		2.0%
attempted by the exact solver		12.1%
of the attempted:	seven-wave Newton	43.8%
	five-wave planar rescue	50.5%
	four-unknown planar solver	2.3%
	degenerate class	1.1%
	HLLD fallback	2.28%
	solved exactly	**97.72%**

Table 7.2: Provenance of the fluxes in a complete 64^2 relativistic-rotor run with the production defaults (solver record, §5a, "Every face of one run").

What "exact" means. An answer is accepted as exact if and only if the state behind it satisfies the full seven-wave jump conditions to 10^{-8}, verified per interface, *and* the fan is admissible: its waves are ordered in space. Two lessons are recorded with this definition. First, a zero residual is not a solution: the jump conditions do not order the waves, and a fan whose Alfvén wave lies on the wrong side of its slow wave can have a perfect residual, hence the wave-order gate. Second, a residual is not a state tolerance: the conversion factor is the condition number of the formulation, so a residual of 10^{-8} pins the state to about 10^{-7} in the planar form but, on about 2% of coplanar faces, only to about 10^{-4} in the seven-wave form, whose Jacobian has a condition number of order 10^4 on those faces against about 30 elsewhere. Re-solving accepted faces at a tolerance of 10^{-12} moved the flux by a median of 10^{-12}–10^{-11} and at most 9.4×10^{-7} (solver record, §5b).

The flux pipeline. A sweep computes the HLLD flux on every face first, then overwrites the faces the exact solver earns (the guide, Part 4):

1. *gates*: superluminal or non-positive states keep HLLD; faces whose relative jump is below 10^{-2} are not attempted and take a linearised flux, which agrees with the exact flux to second order in the jump (Section 9.4); faces whose whole fan lies on one side keep the HLLD flux, which is then the exact upwind flux;
2. *canonical orientation*: each face is replaced by the canonical member of its symmetry orbit (reflection, signed exchanges of the tangential axes, $\boldsymbol{B} \to -\boldsymbol{B}$, with $B_n \geq 0$), solved, and mapped back;
3. *seven-wave Newton* from the network seeds, up to three rounds of 40 iterations; an answer is accepted if it converged, the state on the ray is resolved and physical, and the fan is ordered;
4. *rescues* on the faces it lost: the five-wave planar solver with extra starts, then the four-unknown solver (both only on faces planar to 10^{-6});
5. *fallback*: a face every rung lost takes the HLLD flux (or the linearised flux if its jump is small);
6. the flux is the flux of the state at $\xi = 0$.

Table 7.2 shows where the fluxes of one complete run came from. Two things stand out. Half of the attempted faces are answered by the planar rescue, which reflects the coplanarity of 2D flows. And a few per cent are not answered at all; why is the subject of the next chapter.

Exercises

Exercise 7.1. Verify that the effective enthalpy and internal energy of (7.2) are consistent, $\hat{h} = 1 + \hat{\epsilon} + \hat{p}/\rho$, and derive $\hat{c}_s$.

Solution. $1 + \epsilon + \frac{1}{2}\hat{B}^2\rho + p/\rho + \frac{1}{2}\hat{B}^2\rho = h + \hat{B}^2\rho = h + b^2/\rho$. Along an adiabat of the effective EOS $\hat{B}$ is constant, so $\partial\hat{p}/\partial\rho = hc_s^2 + \hat{B}^2\rho$ and $\hat{c}_s^2 = (hc_s^2 + \hat{B}^2\rho)/\hat{h}$.

Exercise 7.2. Count unknowns and conditions in GR06's method and in the six-unknown formulation. Explain why each slow wave needs a "slack" in the latter but not in the former.

Exercise 7.3. Let L and R be coplanar, with $v_z = B^z = 0$ in both. Show that the reflection $z \to -z$ maps the Riemann problem to itself, and conclude that if the solution is unique it has $v_z = B^z = 0$ everywhere. Which components of $\boldsymbol{u}_6$ are then fixed, and to which values?

Solution. Under $z \to -z$, $v_z \to -v_z$ and $B^z \to -B^z$ while the data are unchanged; a unique solution must be invariant, so $v_z = B^z = 0$. Then $\psi_{\mathrm{CD}} \in \{0, \pi\}$ and $\varphi_L, \varphi_R \in \{0, \pi\}$: three of the six unknowns are discrete, and the remaining continuous problem is the three-unknown planar one.

Exercise 7.4 (Programming)**.** Implement the three-wave solver for $B^x = 0$, $\boldsymbol{v} \cdot \boldsymbol{B} = 0$ by reusing your hydrodynamic solver with the effective EOS (7.2), and compare with the p-method applied directly to the magnetised states.

Chapter 8

Non-regular solutions and non-uniqueness

The construction of Chapter 7 assumes that the solution consists of *regular* waves: Lax shocks of the fast and slow families, rarefactions, rotational discontinuities and a contact. Following Torrilhon (2003a) we call such a solution an *r-solution*, and a solution that contains any other wave (a compound wave or an intermediate shock) a *c-solution*. Because MHD is neither strictly hyperbolic nor convex (Section 3.5), c-solutions are not exotic.

8.1 Non-convexity and compound waves

Where $\nabla_{\boldsymbol{U}} \lambda_k \cdot \boldsymbol{r}_k$ changes sign along a wave curve, the classical construction breaks down: a single shock or a single rarefaction cannot connect the two states admissibly. The admissible structure can then be a *compound wave*, a shock with a rarefaction of the same family attached to it, the shock moving at the characteristic speed of the state it is attached to. Brio and Wu (1988) found that the MHD equations are not convex and observed such a slow compound wave in the numerical solution of their coplanar shock-tube problem, which is now a standard test. Balsara's test 1 (Table 10.1) is its relativistic version; Giacomazzo and Rezzolla (2006) report that its numerical solution develops a left-going slow compound wave (a slow shock adjacent to a slow rarefaction), which their exact solver excludes by construction. In the coplanar geometry a compound wave reverses the tangential field through zero while compressing the gas, which in Newtonian MHD no regular wave can do: a slow shock at most switches the field off, and a reversal at the Alfvén speed is not compressive. For SRMHD, Ibáñez et al. (2015) showed that the convexity of the magnetosonic fields depends on the magnetic field through a generalised fundamental derivative.

8.2 Coplanar non-uniqueness

Barmin et al. (1996) showed that shock-capturing schemes applied to coplanar MHD data produce a non-evolutionary slow compound wave, and that a small out-of-plane perturbation of the field makes a rotational discontinuity split off, after which the compound wave turns into an evolutionary slow shock. Torrilhon (2003b,a) made this precise for Newtonian MHD with an exact solver that includes intermediate waves:

- over a wide range of initial data the r-solution exists and is unique;
- for certain critical data, among them coplanar fields with vanishing transverse velocities, a c-solution exists in addition; the critical data form a hypersurface of codimension one, so arbitrarily close to them the solution is unique again;

- in his coplanar test (states with $\rho, p = 1$ and 0.2, $B_n = 1$, tangential fields at 180°) the r-solution has a single 180° rotation and the c-solution a compound wave instead, and the densities behind the contact differ by about 2%;
- finite-volume schemes (Roe, HLLE, central) show *pseudo-convergence*: on coarse grids they converge towards the c-solution, and only after heavy refinement towards the true one; the effect is strongest when the solution is unique but close to a non-unique one, and is expected for any diffusive scheme.

Takahashi and Yamada (2013) built a solver for regular and non-regular solutions and found, starting from the Brio–Wu data, an initial condition with *no* regular solution, only a non-regular one on the boundary of the regular ones, and showed that non-regular solutions are not unique. Their later exact solver handles all types of intermediate shocks and switch-on/off waves and finds all solutions of a given problem (Takahashi and Yamada, 2014).

8.3 Selection by dissipation

If ideal MHD admits two solutions, something outside ideal MHD must select one. In dissipative MHD the structure of intermediate shocks depends on the dissipation; in Wu's simulations they form from smooth waves and their structures carry free parameters related to their stability (Wu, 1990), and Inoue and Inutsuka (2007) found all MHD shocks evolutionary once dissipation is included. A numerical scheme supplies its own dissipation, and therefore its own selection.

Measured: which of two exact answers a finite-volume code picks The solver record (§5) reports that Balsara's test 1 has two exact solutions in SRMHD, a compound wave and a single intermediate shock, 7.7×10^{-4} apart in the star state. One-dimensional runs with HLLD and HLLE fluxes, piecewise-linear and fifth-order monotonicity-preserving reconstruction (also seventh order and WENO5-Z), at 1024 to 4096 cells, all approach the intermediate-shock solution (to 9.1×10^{-5} at 4096 cells), while the distance to the compound solution stalls near 10^{-3}. Neither the reconstruction order nor the flux function changes the selection. The record's interpretation: the selection happens inside the discontinuity, where every scheme is first order and the profile is set by its dissipation, so a code that reports one of the two answers reports a property of its own solver.

8.4 What this means for an exact solver

An exact solver built on the regular seven-wave family cannot represent compound waves or intermediate shocks. On data whose admissible solution is non-regular it must fail, or it returns a regular answer if one exists. Two further consequences matter in practice.

- *Inadmissible roots exist.* The jump conditions alone admit fans whose waves are in the wrong order; the record (§5, "A zero residual is not a solution") reports that a residual-minimising search converged to such inadmissible roots on 11–12% of interfaces where an admissible root was known. An admissibility (wave-order) check is part of the definition of a solution.
- *Uniqueness within the regular family* is an empirical question for the populations a code meets. On 7620 lanes compared by two independent routes the record found no second admissible regular solution (§5b).

8.5 The failure classes of the reference implementation

The reference implementation keeps a ledger of every interface its exact solver loses. Three classes are documented (solver record, §5–5a and the four-quadrant section of §7):

1. **Compound waves.** Running each lost interface as a one-dimensional shock tube shows a reversal of the tangential field through zero fused to a density and magnitude jump—a structure outside the seven-wave family. On the 64^2 rotor, compound interfaces are 2.5% of the attempted ones and 75.7% of them are unsolvable (2.3% and 68.1% at 128^2). A two-condition rule computed from the states alone,

$$\min(|B_{t,L}|, |B_{t,R}|) < 0.3\,|B_n| \quad \text{and} \quad |\Delta\psi| > \pi/2, \tag{8.1}$$

   (the field is nearly normal to the face and its tangential part turns by more than a right angle) flags 2.5% of the attempted interfaces with 81% precision and 79% recall. This is the regime near $B_t = 0$ where the slow, Alfvén and fast speeds approach one another.

2. **Field nearly normal to the face.** About half of what production lost on the rotor (695 of 1371 failing interfaces in the ledger) is elementary according to the shock-tube test. These interfaces have $|B_n|/|B_t|$ with median 4.4 (0.85 on solved ones), and the Alfvén speed exceeds the sound speed. There the fast waves carry the tangential field and the slow waves carry the pressure, $|B_t|$ is the wrong coordinate for the slow wave, and the planar Newton's residual jumps between branches. The four-unknown solver, which steers such slow waves by the pressure, certifies 135 of 200 of them; with all rescue stages, the exact fraction on the ledger windows rose from 94.39% to 97.86%.

3. **Slow waves riding their Alfvén waves in fast tangential flow.** In a four-quadrant problem with a 45° field (Section 10.5) 93.0% of the attempted faces were solved. The lost faces sit just inside the two quadrants moving at $0.99c$, in flow with $W \approx 6.4$ and magnetisation 0.045, where the slow waves move with their Alfvén waves (gap $\sim 2\times10^{-4}$) and the whole inner fan is about 0.01 wide. Nothing in the solver chain closes 95% of them; the record concludes that a formulation for a slow wave travelling with its Alfvén wave is needed.

Exercises

Exercise 8.1. Explain why critical data of codimension one can make a convergent scheme look as if it converged to the wrong solution (pseudo-convergence), and why refining the grid eventually removes the effect.

Exercise 8.2. Show that the compound-wave rule above is invariant under the symmetries used to canonicalise a face: rotation of the tangential plane about x, $\boldsymbol{B} \to -\boldsymbol{B}$, and the reflection $x \to -x$ (which exchanges L and R and flips the normal components).

Solution. $|B_t|$ and $|B_n|$ are invariant under all three; $\Delta\psi$, the angle between the two tangential fields, is invariant under rotation and under $\boldsymbol{B} \to -\boldsymbol{B}$, and the exchange $L \leftrightarrow R$ only changes its sign.

Exercise 8.3. A finite-volume code converges to the intermediate-shock solution of Balsara's test 1. Should an exact Riemann solver used as its flux return that solution, the compound one, or none? Discuss in terms of consistency between the flux and the dissipation of the scheme.

Chapter 9

Approximate Riemann solvers

An approximate Riemann solver replaces the exact fan by a simpler model that still satisfies the integral form of the conservation law. We derive the HLL solver, then the two relativistic refinements that restore the contact (HLLC) and the Alfvén waves (HLLD), and finally describe linearised solvers.

9.1 HLL

Let $S_L \le S_R$ bound all signal speeds of the Riemann fan, i.e. the fan lies inside $S_L t \le x \le S_R t$. Apply the integral form (1.2) to the rectangle $[S_L T, S_R T] \times [0, T]$: the states on its left and right edges are $\boldsymbol{U}_L$ and $\boldsymbol{U}_R$, so

$$\int_{S_L T}^{S_R T} \boldsymbol{U}(x, T)\,\mathrm{d}x = T(S_R \boldsymbol{U}_R - S_L \boldsymbol{U}_L + \boldsymbol{F}_L - \boldsymbol{F}_R). \tag{9.1}$$

Harten et al. (1983) replace the fan by a single constant state equal to this average,

$$\boldsymbol{U}^{\mathrm{hll}} = \frac{S_R \boldsymbol{U}_R - S_L \boldsymbol{U}_L + \boldsymbol{F}_L - \boldsymbol{F}_R}{S_R - S_L}. \tag{9.2}$$

The flux on the ray $\xi = 0$ follows from the integral form on $[S_L T, 0] \times [0, T]$, which gives $\boldsymbol{F}^{\mathrm{hll}} = \boldsymbol{F}_L + S_L(\boldsymbol{U}^{\mathrm{hll}} - \boldsymbol{U}_L)$, or equivalently (using $[0, S_R T]$) $\boldsymbol{F}^{\mathrm{hll}} = \boldsymbol{F}_R + S_R(\boldsymbol{U}^{\mathrm{hll}} - \boldsymbol{U}_R)$. Eliminating $\boldsymbol{U}^{\mathrm{hll}}$,

$$\hat{\boldsymbol{F}} = \begin{cases} \boldsymbol{F}_L, & S_L \ge 0, \\ \dfrac{S_R \boldsymbol{F}_L - S_L \boldsymbol{F}_R + S_L S_R (\boldsymbol{U}_R - \boldsymbol{U}_L)}{S_R - S_L}, & S_L < 0 < S_R, \\ \boldsymbol{F}_R, & S_R \le 0. \end{cases} \tag{9.3}$$

With $c_{\min} \equiv \max(0, -S_L)$ and $c_{\max} \equiv \max(0, S_R)$ the three cases merge into the single formula used in `hlld.py`,

$$\hat{\boldsymbol{F}} = \frac{c_{\max} \boldsymbol{F}_L + c_{\min} \boldsymbol{F}_R - c_{\max} c_{\min} (\boldsymbol{U}_R - \boldsymbol{U}_L)}{c_{\max} + c_{\min}} \tag{9.4}$$

(identities checked symbolically). The HLL flux is consistent ($\hat{\boldsymbol{F}}(\boldsymbol{U}, \boldsymbol{U}) = \boldsymbol{F}(\boldsymbol{U})$), conservative, and robust: with suitable signal speeds Einfeldt (1988) showed it preserves positivity (the "HLLE" solver). Its weakness is equally clear: by averaging the fan into one state it smears every intermediate wave—contact, Alfvén and slow—over the numerical diffusion of the fastest waves.

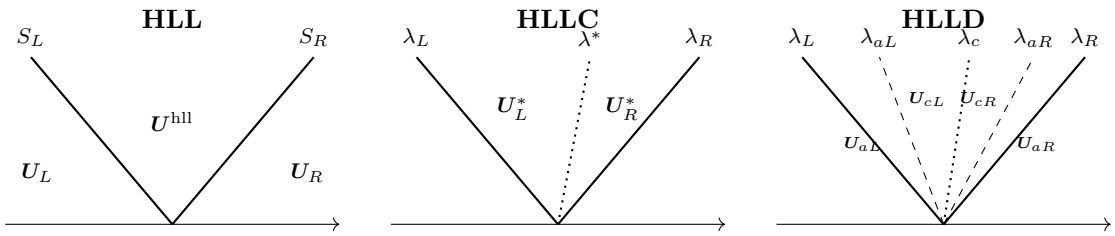


Figure 9.1: The wave models of the HLL, HLLC and HLLD solvers. HLL keeps the two outer waves and one averaged state; HLLC adds the contact; HLLD adds the two Alfvén (rotational) waves, with a single total pressure across the whole fan. None of them models the slow waves.

Signal speeds. Common choices are the minimum and maximum of the outer characteristic speeds of the two states (as in Mignone and Bodo, 2006), from the quartic (3.15), or the isotropic bound (3.18) (Gammie et al., 2003), which by Proposition 3.3 can only overestimate them. The code uses the latter.

9.2 HLLC for relativistic MHD

Mignone and Bodo (2006) restore the contact. The fan is modelled by two outer waves λ_L, λ_R and a contact λ^* separating two star states $\boldsymbol{U}_L^*, \boldsymbol{U}_R^*$, related by the RH conditions across each wave and constrained by the two consistency conditions that the star states average to $\boldsymbol{U}^{\mathrm{hll}}$ and the star fluxes to $\boldsymbol{F}^{\mathrm{hll}}$. The contact moves with the fluid, $\lambda^* = v_x^*$, and the total pressure and normal velocity are continuous across it.

The case $B^x = 0$. The star states have $m_x^* = (E^* + p^*)v^*$ and star fluxes $F_{m_x}^* = m_x^* v^* + p^*$, $F_E^* = m_x^*$, with common p^*, v^* (p is the total pressure here). Hence on both sides

$$F_{m_x}^* - v^* F_E^* = p^*, \qquad m_x^* - v^* E^* = p^* v^*. \tag{9.5}$$

Both relations are linear in the star quantities. The consistency conditions express $\boldsymbol{U}^{\mathrm{hll}}$ and $\boldsymbol{F}^{\mathrm{hll}}$ as averages of the two star values with weights that sum to one, so the relations hold for the HLL averages too: $p^* = F_{m_x}^{\mathrm{hll}} - v^* F_E^{\mathrm{hll}}$ and $m_x^{\mathrm{hll}} = (E^{\mathrm{hll}} + p^*)v^*$. Eliminating p^*:

$$F_E^{\mathrm{hll}} (v^*)^2 - (E^{\mathrm{hll}} + F_{m_x}^{\mathrm{hll}})v^* + m_x^{\mathrm{hll}} = 0, \tag{9.6}$$

the root with the minus sign being the physical one (Mignone and Bodo, 2006, eq. 47).

The case $B^x \neq 0$. Now $\boldsymbol{v}$, $\boldsymbol{B}$ and the total pressure are continuous at the contact. The tangential components of the consistency conditions give

$$\boldsymbol{B}_t^* = \boldsymbol{B}_t^{\mathrm{hll}}, \qquad B^x v_t^* = B_t^* v_x^* - F_{B_t}^{\mathrm{hll}}, \tag{9.7}$$

and the normal-momentum and energy components give (Mignone and Bodo, 2006, eqs. 39–40)

$$E^{\mathrm{hll}} v_x^* + p^* v_x^* - B^x(\boldsymbol{v}^* \cdot \boldsymbol{B}^*) = m_x^{\mathrm{hll}}, \tag{9.8}$$

$$[F_E^{\mathrm{hll}} - B^x(\boldsymbol{v}^* \cdot \boldsymbol{B}^*)]v_x^* - \left(\frac{B^x}{\gamma^*}\right)^2 + p^* - F_{m_x}^{\mathrm{hll}} = 0, \tag{9.9}$$

with γ^* the star Lorentz factor. Multiplying the second by v_x^* and subtracting the first gives a quadratic $a(v_x^*)^2 + b v_x^* + c = 0$ with

$$a = F_E^{\mathrm{hll}} - \boldsymbol{B}_\perp^{\mathrm{hll}} \cdot \boldsymbol{F}_{B_\perp}^{\mathrm{hll}}, \quad b = -F_{m_x}^{\mathrm{hll}} - E^{\mathrm{hll}} + |\boldsymbol{B}_\perp^{\mathrm{hll}}|^2 + |\boldsymbol{F}_{B_\perp}^{\mathrm{hll}}|^2, \quad c = m_x^{\mathrm{hll}} - \boldsymbol{B}_\perp^{\mathrm{hll}} \cdot \boldsymbol{F}_{B_\perp}^{\mathrm{hll}}, \tag{9.10}$$

again taking the minus root; v_t^* and p^* follow, and the conserved star states from the RH conditions across the outer waves. Two remarks of Mignone and Bodo (2006) deserve emphasis: the formulas for $B^x \neq 0$ do not reduce continuously to those for $B^x = 0$, because the contact becomes a tangential discontinuity; and the transverse velocities are ill-defined as $B^x \to 0$, with a possible $O(1/B^x)$ term in the transverse momenta that vanishes for strictly two-dimensional flows ($B^z = v^z = 0$) but not in general.

9.3 HLLD for relativistic MHD

Mignone et al. (2009) extend the multi-state approach of Miyoshi and Kusano (2005) to SRMHD. The model fan has five waves—fast λ_L, Alfvén λ_{aL}, contact λ_c, Alfvén λ_{aR}, fast λ_R—and four intermediate states $\boldsymbol{U}_{aL}, \boldsymbol{U}_{cL}, \boldsymbol{U}_{cR}, \boldsymbol{U}_{aR}$; slow waves are not modelled, so the total pressure p is the same everywhere in the fan. The single unknown is p.

Across the fast waves. With $\boldsymbol{R} \equiv \lambda \boldsymbol{U} - \boldsymbol{F}$ evaluated on the outer state (known; $\lambda = \lambda_L$ or λ_R), the jump conditions for the state behind the fast wave read (Mignone et al., 2009, eqs. 17–20)

$$(\lambda - v^x) D = R_D, \tag{9.11}$$

$$(\lambda - v^x) w W^2 v^k + b^k (b^x - \lambda b^0) - p\,\delta_{kx} = R_{m_k}, \tag{9.12}$$

$$(\lambda - v^x) w W^2 - \lambda p + b^0 (b^x - \lambda b^0) = R_E, \tag{9.13}$$

$$(\lambda - v^x) B^k + B^x v^k = R_{B^k}, \tag{9.14}$$

with w the total enthalpy and p the total pressure. The last gives $B^k = (R_{B^k} - B^x v^k)/(\lambda - v^x)$, and combining momentum and energy gives the velocity as an explicit function of p (their eqs. 23–30; the auxiliary quantities $\tilde{A}, \ldots, \tilde{X}$ are unrelated to those of Chapter 3):

$$v^x = \frac{B^x(\tilde{A} B^x + \lambda \tilde{C}) - (\tilde{A} + \tilde{G})(p + R_{m_x})}{\tilde{X}}, \qquad v^t = \frac{\tilde{Q} R_{m_t} + R_{B^t}\left[\tilde{C} + B^x(\lambda R_{m_x} - R_E)\right]}{\tilde{X}}, \tag{9.15}$$

$$\tilde{A} = R_{m_x} - \lambda R_E + p(1 - \lambda^2), \quad \tilde{G} = |\boldsymbol{R}_{B_t}|^2, \quad \tilde{C} = \boldsymbol{R}_{m_t} \cdot \boldsymbol{R}_{B_t}, \tag{9.16}$$

$$\tilde{Q} = -\tilde{A} - \tilde{G} + (B^x)^2(1 - \lambda^2), \quad \tilde{X} = B^x(\tilde{A}\lambda B^x + \tilde{C}) - (\tilde{A} + \tilde{G})(\lambda p + R_E), \tag{9.17}$$

and the total enthalpy is $w = p + (R_E - \boldsymbol{v} \cdot \boldsymbol{R}_m)/(\lambda - v^x)$ (their eq. 31).

Across the Alfvén waves. The invariant of Section 4.5 is the key: $\boldsymbol{K} \equiv \boldsymbol{\sigma}/\sigma^0$, with $\sigma^\mu = \eta u^\mu + b^\mu$ and $\eta = \pm\,\mathrm{sign}(B^x)\sqrt{w}$ (+ on the right, − on the left), is continuous across the Alfvén wave, and $K^x = \lambda_a$. Expressed through p and the outer state (Mignone et al., 2009, eq. 43),

$$K^k = \frac{R_{m_k} + p\,\delta_{kx} + R_{B^k}\,\eta}{\lambda p + R_E + B^x \eta}. \tag{9.18}$$

At the contact. $\boldsymbol{B}$ is continuous; the consistency condition between the two Alfvén waves gives

$$\boldsymbol{B}_c = \frac{\left[\boldsymbol{B}(\lambda_a - v^x) + B^x \boldsymbol{v}\right]_{aR} - \left[\boldsymbol{B}(\lambda_a - v^x) + B^x \boldsymbol{v}\right]_{aL}}{\lambda_{aR} - \lambda_{aL}}, \tag{9.19}$$

and inverting the definition of $\boldsymbol{K}$ on each side gives the velocity at the contact, $\boldsymbol{v}_c = \boldsymbol{K} - \boldsymbol{B}_c\,(1 - K^2)/(\eta - \boldsymbol{K} \cdot \boldsymbol{B}_c)$ (their eq. 47). Continuity of the normal velocity is the equation for

p:

$$\boxed{f(p) \equiv v^x_{cR}(p) - v^x_{cL}(p) = \Delta K^x - B^x\Big[\frac{1-K_R^2}{\eta_R - \boldsymbol{K}_R\cdot\boldsymbol{B}_c} - \frac{1-K_L^2}{\eta_L - \boldsymbol{K}_L\cdot\boldsymbol{B}_c}\Big] = 0,} \tag{9.20}$$

with $\Delta K^x = K^x_R - K^x_L$. Multiplying out, this is eq. (48) of Mignone et al. (2009), $\Delta K^x[1 - B^x(Y_R - Y_L)] = 0$ with $Y_S = (1-K_S^2)/(\eta_S \Delta K^x - \boldsymbol{K}_S\cdot\Delta K^x \boldsymbol{B}_c)$.

Solution and safeguards. MUB09 solve (9.20) by the secant method from the HLL total pressure, or from the solution of the $B^x = 0$ limit when $(B^x)^2/p^{\rm hll} < 0.1$. They warn that (9.20) may have several roots and impose $w_L > p$, $w_R > p$ and the ordering $v^x_{aL} > \lambda_L$, $v^x_{cL} > \lambda_{aL}$, $v^x_{aR} < \lambda_R$, $v^x_{cR} < \lambda_{aR}$; where these fail, they switch to HLL, which in their tests happened in usually less than 0.1% of the zones. In the limit $B^x \to 0$ the equation for p becomes a quadratic and the same algorithm applies.

In the code: HLLD

`HLLD/src/physics/hlld.py` follows MUB09: `HLLDComputation` evaluates the quantities above for a trial p and returns $-f(p)$, i.e. the same equation with the opposite sign. The start is MUB09's rule (from the total pressure of the HLL state, or from the $B_x = 0$ limit), the two starting points are moved apart until f changes sign (at most five times), and a clamped secant iterates until $|f(p)| < 10^{-10}$ (about 3.5 iterations according to the guide, §2.3; Newton variants were measured to be slower and less reliable because the residual has several roots, and a comment in the code records an interface with seven sign changes of f on $10^{-3} \le p \le 10^2$, where the starting point decides the root). Faces fall back to HLLE when the root-find fails, when the wave order is violated, or when $B_n = 0$ on both sides; the last is a difference from MUB09, whose formulation covers $B^x \to 0$, and it matters in 2D because a field initially along x makes every y-sweep degenerate.

9.4 Roe-type and linearised solvers

A linearised solver decomposes the jump into the characteristic fields of an averaged state,

$$\boldsymbol{U}_R - \boldsymbol{U}_L = \sum_k \alpha_k \boldsymbol{r}_k, \qquad \hat{\boldsymbol{F}} = \tfrac{1}{2}(\boldsymbol{F}_L + \boldsymbol{F}_R) - \tfrac{1}{2}\sum_k |\lambda_k|\,\alpha_k \boldsymbol{r}_k, \tag{9.21}$$

which resolves every wave family, including contacts and Alfvén waves, as a discontinuity. It needs the full eigenstructure, which in SRMHD has no closed-form Roe average, and the eigenvectors must be renormalised at degeneracies (Section 3.5). Komissarov (1999) built a Godunov-type scheme for RMHD on linear Riemann solvers; Balsara (2001) a total-variation-diminishing scheme; and Antón et al. (2010) a full-wave-decomposition solver based on their renormalised eigenvectors, which they found less diffusive than HLL and HLLC and comparable in accuracy to HLLD, but more expensive in one dimension, with the relative cost improving in multidimensional runs.

In the code: the linearised flux below the weak-jump gate

`HLLD/src/physics/linearised.py` uses a linearisation for faces whose relative jump is below 10^{-2}: the Jacobian $(\partial \boldsymbol{U}/\partial \boldsymbol{V})^{-1}\partial \boldsymbol{F}/\partial \boldsymbol{V}$ at the arithmetic mean state (central differences), its eigen-decomposition, and the state on the ray $\xi = 0$, $\boldsymbol{V}^* = \boldsymbol{V}_L + \sum_{\lambda_k<0}\alpha_k \boldsymbol{r}_k$; the flux is $\boldsymbol{F}(\boldsymbol{V}^*)$. For a weak jump this agrees with the exact solution to second order in the jump, whereas HLLD's error is first order there. It is never counted as exact.

9.5 Trade-offs

Mattia and Mignone (2022) compared five solvers that avoid the full eigenstructure (HLL, HLLC, HLLD, HLLEM and GFORCE) on standard one- and multidimensional tests. For weak to moderate magnetisation HLLD was the most accurate, followed by HLLC; GFORCE was less dissipative than HLL and equally robust at strong magnetisation, and HLLEM did not justify its cost. MUB09 found that HLL needed up to about twice the resolution of HLLD to match its results in their multidimensional tests. The exact solver is the end point of this sequence. In the reference implementation's 64^2 rotor runs, $L_1(\rho)$ against a 512^2 HLLD reference at $t = 0.4$ is 0.2881 with HLLD and 0.2874 with the exact flux on the attempted faces (solver record, §5a), and changing the reconstruction from piecewise-linear to fifth order moved the rotor some 24 times more than switching on the exact flux (§7). On the magnetised four-quadrant problem at 64^2, by contrast, the exact flux reduced $L_1(\rho)$ against the 512^2 reference at $t = 0.4$ from 0.1100 (HLLD) to 0.1019, about 7% of the gap. Which solver is "best" is therefore problem- and resolution-dependent; what the exact solver provides unambiguously is a reference.

Exercises

Exercise 9.1. Derive (9.3) and show that $\hat{\boldsymbol{F}}$ is continuous when S_L or S_R passes through zero.

Exercise 9.2. Show that, for an isolated stationary contact ($\boldsymbol{v}_L = \boldsymbol{v}_R = 0$, $p_L = p_R$, $\boldsymbol{B} = 0$, $\rho_L \neq \rho_R$), the HLL flux has a nonzero mass component while the HLLC flux vanishes.

Solution. $F_D = 0$ on both sides but $\boldsymbol{U}_R - \boldsymbol{U}_L \neq 0$, so the HLL mass flux is $-S_L S_R (D_R - D_L)/(S_R - S_L) \neq 0$. In HLLC, $v^* = 0$ (the quadratic has $c = m_x^{\mathrm{hll}} = 0$ and the minus root vanishes), so the star states carry no mass flux.

Exercise 9.3. Derive (9.6) from the two consistency conditions, filling in the steps above.

Exercise 9.4. Show that (9.20) and MUB09's form $\Delta K^x [1 - B^x (Y_R - Y_L)] = 0$ are the same equation, and check that the code's residual is $-f(p)$.

Chapter 10

Riemann solvers in multidimensional codes

10.1 The finite-volume update

In two dimensions the semi-discrete finite-volume scheme for cell averages $\boldsymbol{U}_{ij}$ is

$$\frac{\mathrm{d}\boldsymbol{U}_{ij}}{\mathrm{d}t} = -\frac{\hat{\boldsymbol{F}}_{i+1/2,j} - \hat{\boldsymbol{F}}_{i-1/2,j}}{\Delta x} - \frac{\hat{\boldsymbol{G}}_{i,j+1/2} - \hat{\boldsymbol{G}}_{i,j-1/2}}{\Delta y} \equiv \mathcal{L}(\boldsymbol{U})_{ij}, \tag{10.1}$$

where each face flux is obtained from a one-dimensional Riemann problem in the direction normal to the face, with the left and right states reconstructed at the face. For a y-face the components are permuted so that y plays the role of x; the normal field B^y is then the constant parameter of that Riemann problem. The scheme is *unsplit* when both flux differences are evaluated from the same state, as in (10.1).

In the code: sweeps

In `HLLD/src/physics/` a sweep computes all face fluxes in one direction (`sweep.py`). The flux functions take the direction as a parameter and cycle the tangential components, $t_1 = (\texttt{idir}+1) \bmod 3$, $t_2 = (\texttt{idir}+2) \bmod 3$; the exact solver works in its own frame, into which `to_solver_frame` rotates every face.

10.2 Reconstruction

Godunov's scheme is first order because the data are piecewise constant. Second order in smooth regions is obtained by piecewise-linear reconstruction with a limited slope. With the monotonised-central (MC) limiter of van Leer (1977),

$$\Delta_i = \operatorname{minmod}\Big(2(q_i - q_{i-1}),\ \tfrac{1}{2}(q_{i+1} - q_{i-1}),\ 2(q_{i+1} - q_i)\Big), \tag{10.2}$$

$$q^{L}_{i+1/2} = q_i + \tfrac{1}{2}\Delta_i, \qquad q^{R}_{i-1/2} = q_i - \tfrac{1}{2}\Delta_i, \tag{10.3}$$

where minmod returns the argument of smallest modulus if all have the same sign and zero otherwise. Higher orders use weighted essentially non-oscillatory (WENO) reconstruction, which combines three third-order candidate stencils with nonlinear weights based on smoothness indicators (Jiang and Shu, 1996); the WENO-Z variant modifies the weights with a global smoothness indicator to reduce dissipation near critical points (Borges et al., 2008). Reconstruction acts on primitive variables. In relativity it is safer to reconstruct the spatial four-velocity $W\boldsymbol{v}$ than $\boldsymbol{v}$, since any reconstructed $W\boldsymbol{v}$ gives a subluminal velocity.

In the code: reconstruction

`reconstruction.py` offers `pcm`, `minmod`, `mc` (production), `weno5z`, `mp5` and `mp7`; it reconstructs $W\boldsymbol{v}$, and a face whose reconstructed state violates the density or pressure floor falls back to piecewise-constant data (the guide, §2.2).

10.3 Time integration

The method of lines integrates $\mathrm{d}\boldsymbol{U}/\mathrm{d}t = \mathcal{L}(\boldsymbol{U})$ with a strong-stability-preserving Runge–Kutta scheme; the third-order scheme of Shu and Osher (1988) is

$$\begin{aligned} \boldsymbol{U}^{(1)} &= \boldsymbol{U}^n + \Delta t\, \mathcal{L}(\boldsymbol{U}^n), \\ \boldsymbol{U}^{(2)} &= \tfrac{3}{4}\boldsymbol{U}^n + \tfrac{1}{4}\left[\boldsymbol{U}^{(1)} + \Delta t\, \mathcal{L}(\boldsymbol{U}^{(1)})\right], \\ \boldsymbol{U}^{n+1} &= \tfrac{1}{3}\boldsymbol{U}^n + \tfrac{2}{3}\left[\boldsymbol{U}^{(2)} + \Delta t\, \mathcal{L}(\boldsymbol{U}^{(2)})\right]. \end{aligned} \tag{10.4}$$

Each stage needs the primitive variables of every cell (Section 2.6), a reconstruction, and a Riemann solve on every face. The time step obeys a CFL condition with the largest signal speed; the reference code uses this scheme with Courant number 0.25 (the guide, §2.5).

10.4 Constrained transport

The constraint $\nabla \cdot \boldsymbol{B} = 0$ is preserved by the continuous equations but not, in general, by a cell-centred finite-volume update. Constrained transport (Evans and Hawley, 1988) stores the normal field components on the faces, $B^x_{i-1/2,j}$ and $B^y_{i,j-1/2}$, and updates them by Stokes' theorem from the electric field $E_z = v_y B_x - v_x B_y$ (from $\boldsymbol{E} = -\boldsymbol{v} \times \boldsymbol{B}$) at the cell corners:

$$\partial_t B^x_{i-1/2,j} = -\frac{E_{z,i-1/2,j+1/2} - E_{z,i-1/2,j-1/2}}{\Delta y}, \tag{10.5}$$

$$\partial_t B^y_{i,j-1/2} = +\frac{E_{z,i+1/2,j-1/2} - E_{z,i-1/2,j-1/2}}{\Delta x}. \tag{10.6}$$

The discrete divergence of cell (i,j), $(B^x_{i+1/2,j} - B^x_{i-1/2,j})/\Delta x + (B^y_{i,j+1/2} - B^y_{i,j-1/2})/\Delta y$, has a time derivative in which each of the four corner values appears twice with opposite signs, so it is conserved to round-off whatever the corner values are (Exercise 10.1).

The Riemann solver supplies the EMF. The induction fluxes of (2.13) are

$$F^x(B^y) = v_x B^y - v_y B^x = -E_z, \qquad F^y(B^x) = v_y B^x - v_x B^y = +E_z, \tag{10.7}$$

so every face flux carries a face-centred value of E_z, upwinded by the Riemann solver in the direction normal to the face. Constrained transport needs it at the corners, where it must be upwinded in both directions. The arithmetic average of the four face values (Balsara and Spicer, 1999) lacks the upwinding along the face. Gardiner and Stone (2005) derived corner EMFs that add corrections built from the cell-centred E_z, chosen by the sign of the mass flux through the adjacent faces, and that reduce to the one-dimensional solver's flux for grid-aligned flow. Mignone and Del Zanna (2021) give a systematic construction of upwind constrained-transport schemes from the information of the Riemann solver and compare the alternatives.

test	state	ρ	p	v_x	v_y	v_z	B_x	B_y	B_z	t
1 ($\Gamma = 2$)	L	1.0	1.0	0.0	0.0	0.0	0.5	1.0	0.0	0.4
	R	0.125	0.1	0.0	0.0	0.0	0.5	-1.0	0.0	
2	L	1.0	30.0	0.0	0.0	0.0	5.0	6.0	6.0	0.4
	R	1.0	1.0	0.0	0.0	0.0	5.0	0.7	0.7	
3	L	1.0	1000.0	0.0	0.0	0.0	10.0	7.0	7.0	0.4
	R	1.0	0.1	0.0	0.0	0.0	10.0	0.7	0.7	
4	L	1.0	0.1	0.999	0.0	0.0	10.0	7.0	7.0	0.4
	R	1.0	0.1	-0.999	0.0	0.0	10.0	-7.0	-7.0	
5	L	1.08	0.95	0.4	0.3	0.2	2.0	0.3	0.3	0.55
	R	1.0	1.0	-0.45	-0.2	0.2	2.0	-0.7	0.5	

Table 10.1: The relativistic MHD Riemann problems of Balsara (2001, Table 1) (p is the gas pressure, t the final time). Test 1 uses $\Gamma = 2$; the others $\Gamma = 1.6666$ (as printed; 5/3 in Giacomazzo and Rezzolla (2006)). The original runs use a unit domain of 1600 zones with the discontinuity at its centre. Verified against Balsara's table; the same values are in Giacomazzo and Rezzolla (2006, Table 4) and in `rmhd/initial_data.py` (cases 8–12).

In the code: the corner EMF

`ct.py` implements the Gardiner–Stone corner EMF. For the corner $(i-\frac{1}{2}, j-\frac{1}{2})$, with face EMFs E^y_E, E^y_W on the y-faces to its east and west and E^x_N, E^x_S on the x-faces to its north and south, cell-centred values $E_{NE}, E_{NW}, E_{SE}, E_{SW}$, and upwind signs s of the mass flux on those faces,

$$\begin{aligned} E_z = {}& \tfrac{1}{4}(E^y_E + E^y_W + E^x_N + E^x_S) \\ & + \tfrac{1}{8}\Big[(1-s_S)(E^y_E - E_{SE}) + (1+s_S)(E^y_W - E_{SW}) \\ & \qquad - (1-s_N)(E_{NE} - E^y_E) - (1+s_N)(E_{NW} - E^y_W)\Big] \\ & + \tfrac{1}{8}\Big[(1-s_W)(E^x_N - E_{NW}) + (1+s_W)(E^x_S - E_{SW}) \\ & \qquad - (1-s_E)(E_{NE} - E^x_N) - (1+s_E)(E_{SE} - E^x_S)\Big], \end{aligned}$$

where s_N, s_S are the signs on the north and south x-faces and s_E, s_W on the east and west y-faces. Dropping the bracketed corrections (option `upwind=False`) gives the four-face average; with $s = 0$ the corrections are the centred average of the two one-sided estimates. The code sets $s = 0$ in a deadband around zero mass flux: in a static medium the computed mass flux is round-off with no definite sign, and taking its sign broke the discrete symmetry of the rotor. The measured $|\nabla \cdot \boldsymbol{B}|$ stays at $\sim 10^{-13}$ (the guide, §2.4).

10.5 Standard tests

One-dimensional shock tubes. Table 10.1 lists the five Riemann problems of Balsara (2001), which, with those of Komissarov (1999) (some of whose parameters were corrected in Komissarov (2002)), are the standard SRMHD shock tubes. Giacomazzo and Rezzolla (2006) give exact solutions for all five.

The relativistic rotor. Del Zanna et al. (2003) adapted the classical MHD rotor to SRMHD: a disk of radius 0.1 with $\rho = 10$ rotating rigidly with $\omega = 9.95$ (rim speed 0.995, $W_{\max} \approx 10$) inside a static medium with $\rho = 1$; uniform $p = 1$ and $\boldsymbol{B} = (1, 0, 0)$; $\Gamma = 5/3$; output at $t = 0.4$. The field winds up and brakes the rotor, launching fast and Alfvén waves; the central density is

swept into a shell. The problem is invariant under a rotation by π, which gives a free measure of numerical error.

The relativistic Orszag–Tang vortex. The Newtonian vortex of Orszag and Tang (1979) has several relativistic versions. van der Holst et al. (2008) use the doubly periodic domain $[0,2\pi]^2$ with $\rho=1$, $p=10$, $\Gamma=4/3$, $\boldsymbol{v}=(-A\sin y,\,A\sin x)$ with $A=0.99/\sqrt{2}$ (maximum $W\approx 7$) and $\boldsymbol{B}=(-\sin y,\sin 2x)$. Beckwith and Stone (2011) use the unit square with $\rho=25/(36\pi)$, $p=5/(12\pi)$, $\Gamma=5/3$, velocity amplitude 0.5 per component (printed as $V^i=[0.5\sin(2\pi y),0.5\sin(2\pi x)]$; the usual orientation has $v_x=-0.5\sin 2\pi y$), and $A_z=(B_0/4\pi)\cos 4\pi x+(B_0/2\pi)\cos 2\pi y$ with $B_0=1/\sqrt{4\pi}$, shown at $t=1$. The reference code's problem combines the flow of the former (peak speed 0.99, $\Gamma=4/3$) with the density, pressure and field of the latter, as its documentation states.

Four-quadrant problems. Del Zanna and Bucciantini (2002) proposed a relativistic two-dimensional Riemann problem on a square divided into quadrants with $(\rho,v_x,v_y,p)=(0.1,0,0,0.01)$ (NE), $(0.1,0.99,0,1)$ (NW), $(0.5,0,0,1)$ (SW) and $(0.1,0,0.99,1)$ (SE), evolved to $t=0.4$; it produces two shocks and two shear layers. The reference implementation adds a uniform in-plane field $\boldsymbol{B}=B_0(\cos 45^\circ,\sin 45^\circ,0)$, $B_0=0.5$, with $\Gamma=5/3$ on $[-0.5,0.5]^2$; its documentation records that no published magnetised version was found among the codes it checked.

Exercises

Exercise 10.1. Show that the constrained-transport update preserves the discrete divergence of every cell exactly, for arbitrary corner values of E_z.

Solution. Label the corners of cell (i,j) NE, NW, SE, SW. The update gives $\partial_t B^x_{i+1/2,j}=-(E_{NE}-E_{SE})/\Delta y$, $\partial_t B^x_{i-1/2,j}=-(E_{NW}-E_{SW})/\Delta y$, $\partial_t B^y_{i,j+1/2}=(E_{NE}-E_{NW})/\Delta x$, $\partial_t B^y_{i,j-1/2}=(E_{SE}-E_{SW})/\Delta x$. Hence $\Delta x\Delta y\,\partial_t(\nabla\cdot\boldsymbol{B})_{ij}=-(E_{NE}-E_{SE})+(E_{NW}-E_{SW})+(E_{NE}-E_{NW})-(E_{SE}-E_{SW})=0$.

Exercise 10.2. Show that $F^x(B^y)=-E_z$ and $F^y(B^x)=+E_z$ with $\boldsymbol{E}=-\boldsymbol{v}\times\boldsymbol{B}$.

Exercise 10.3. For Balsara's test 1, compute the magnetic and gas pressures and the magnetisation $\sigma=b^2/(\rho h)$ in both initial states, and identify the magnetically dominated side.

Solution. With $\boldsymbol{v}=0$, $b^2=B^2=1.25$ on both sides, so $b^2/2=0.625$. For $\Gamma=2$, $h=1+2p/\rho$: $h_L=3$, $\sigma_L=1.25/3\approx 0.42$; $h_R=2.6$, $\rho_R h_R=0.325$, $\sigma_R\approx 3.8$. The right state is magnetically dominated (gas pressure 0.1 against magnetic pressure 0.625).

Exercise 10.4. Show that the MC slope equals the centred slope $\tfrac{1}{2}(q_{i+1}-q_{i-1})$ where the data are smooth and monotone and the two one-sided slopes do not differ by more than a factor of three, and that it vanishes at local extrema.

Appendix A

Algebra of b^μ and the components of $T^{\mu\nu}$

With $u^\mu = W(1, \boldsymbol{v})$, $\eta = \boldsymbol{v} \cdot \boldsymbol{B}$ and $w = \rho h + b^2$:

$$
\begin{aligned}
&b^0 = W\eta = \boldsymbol{v} \cdot \boldsymbol{b}, && b^i = \frac{B^i}{W} + W\eta\, v^i, && b_0 = -W\eta,\\
&b^2 = \frac{B^2}{W^2} + \eta^2, && W^2 b^2 - (b^0)^2 = B^2, && B^i = u^0 b^i - u^i b^0 = W(b^i - b^0 v^i),\\
&u_\mu b^\mu = 0, && {}^*F^{\mu\nu} = u^\mu b^\nu - u^\nu b^\mu, && \boldsymbol{E} = -\boldsymbol{v} \times \boldsymbol{B}.
\end{aligned}
$$

Components of $T^{\mu\nu} = w\, u^\mu u^\nu + P g^{\mu\nu} - b^\mu b^\nu$:

$$
\begin{aligned}
T^{00} &= wW^2 - P - (b^0)^2 = \rho h W^2 + B^2 - p - \tfrac{1}{2}b^2 = \rho h W^2 - p + \tfrac{1}{2}(E^2 + B^2),\\
T^{0i} &= wW^2 v^i - b^0 b^i = (\rho h W^2 + B^2) v^i - \eta B^i = \rho h W^2 v^i + (\boldsymbol{E} \times \boldsymbol{B})^i,\\
T^{ij} &= wW^2 v^i v^j + P\delta^{ij} - b^i b^j = (\rho h W^2 + B^2) v^i v^j - \eta(B^i v^j + B^j v^i) - \frac{B^i B^j}{W^2} + P\delta^{ij}\\
&= S^j v^i + P\delta^{ij} - \frac{B^i b^j}{W},
\end{aligned}
$$

with $E^2 = v^2 B^2 - \eta^2$. In the fluid rest frame $b^\mu = (0, \boldsymbol{B}')$, $T^{00} = e + \frac{1}{2}b^2$, $T^{0i} = 0$ and $T^{ij} = (p + \frac{1}{2}b^2)\delta^{ij} - b^i b^j$. Useful combinations: $h_t = h + b^2/\rho$ (total specific enthalpy), $w = \rho h_t$, $c_a^2 = b^2/w$, $\zeta^2 = c_s^2 + c_a^2 - c_s^2 c_a^2$, $\sigma = b^2/(\rho h) = c_a^2/(1 - c_a^2)$. All identities in this appendix were checked symbolically (Chapter D).

Appendix B

Eigenstructure

B.1 Characteristic matrix in covariant variables

Following Anile (1989) and Antón et al. (2010), take the ten covariant variables $\tilde{\boldsymbol{U}} = (u^\mu, b^\mu, p, s)$, s the specific entropy. For a characteristic covector ϕ_μ, with $a = u^\mu\phi_\mu$, $\mathcal{B} = b^\mu\phi_\mu$, $G = \phi^\mu\phi_\mu$ and $\mathcal{E} \equiv w$, the characteristic matrix acting on $(\delta u^\nu, \delta b^\nu, \delta p, \delta s)$ is

$$\mathcal{A}^\alpha\phi_\alpha = \begin{pmatrix} \mathcal{E}a\,\delta^\mu_\nu & m^\mu{}_\nu & l^\mu & 0^\mu \\ \mathcal{B}\,\delta^\mu_\nu & -a\,\delta^\mu_\nu & f^\mu & 0^\mu \\ \rho h\,\phi_\nu & 0_\nu & a/c_s^2 & 0 \\ 0_\nu & 0_\nu & 0 & a \end{pmatrix}, \tag{B.1}$$

with

$$m^\mu{}_\nu = (\phi^\mu + 2au^\mu)b_\nu - \mathcal{B}\,\delta^\mu_\nu, \qquad l^\mu = \phi^\mu + \frac{(\rho h - b^2/c_s^2)\,a\,u^\mu + \mathcal{B}\,b^\mu}{\rho h},$$
$$f^\mu = \frac{1}{\rho h}\Big(\frac{a}{c_s^2}b^\mu - \mathcal{B}u^\mu\Big).$$

Its determinant is

$$\det(\mathcal{A}^\alpha\phi_\alpha) = \mathcal{E}\,a^2\mathcal{A}^2\mathcal{N}_4, \qquad \mathcal{A} = \mathcal{E}a^2 - \mathcal{B}^2, \qquad \mathcal{N}_4 = \rho h\Big(\frac{1}{c_s^2} - 1\Big)a^4 - \Big(\rho h + \frac{b^2}{c_s^2}\Big)a^2G + \mathcal{B}^2G. \tag{B.2}$$

Remark B.1 (A sign in the printed matrix). In Antón et al. (2010) the (b, b) block is $-u^\mu\delta^\alpha_\beta$ in the general form (their eq. 22), i.e. $-a\,\delta^\mu_\nu$ after contraction with ϕ, but $+a\,\delta^\mu_\nu$ in the contracted form (their eq. 32); and l^μ includes the term $\mathcal{B}b^\mu/(\rho h)$ in the text below eq. 32 but not in their eq. 26. We checked numerically, at an arbitrary state and several values of λ, that with $-a\,\delta^\mu_\nu$ and with the $\mathcal{B}b^\mu/(\rho h)$ term the determinant identity above holds exactly and the eigenvectors below are exact null vectors; with $+a\,\delta^\mu_\nu$ neither holds (Chapter D).

B.2 Right eigenvectors

From Antón et al. (2010, eqs. 46–50), who take them from Anile (1989):

$$\boldsymbol{r}_e = (0^\alpha, 0^\alpha, 0, 1)^\mathsf{T}, \tag{B.3}$$
$$\boldsymbol{r}_{a,\pm} = \big(a\,\varepsilon^\alpha,\ \mathcal{B}\,\varepsilon^\alpha,\ 0,\ 0\big)^\mathsf{T}, \qquad \varepsilon^\alpha = \varepsilon^\alpha{}_{\beta\gamma\delta}u^\beta\phi^\gamma b^\delta, \tag{B.4}$$
$$\boldsymbol{r}_{m,\pm} = \big(a\,d^\alpha,\ \mathcal{B}\,d^\alpha + a\mathcal{A}f^\alpha,\ a^2\mathcal{A},\ 0\big)^\mathsf{T}, \tag{B.5}$$

for the entropy, Alfvén and magnetosonic (m = s, f) waves, with

$$d^\alpha = \frac{c_s^2 - 1}{c_s^2}\Big(\frac{a^4}{G}(\phi^\alpha + a u^\alpha) - \frac{a^2 \mathcal{B}}{\rho h} b^\alpha\Big), \qquad f^\alpha = \frac{1}{\rho h}\Big(\frac{a}{c_s^2} b^\alpha - \mathcal{B} u^\alpha\Big), \tag{B.6}$$

each evaluated at the corresponding root ($a = 0$; $\mathcal{A} = 0$; $\mathcal{N}_4 = 0$). At the Alfvén roots $\mathcal{B}/a = \mp\sqrt{\mathcal{E}}$.

Renormalisation. The Alfvén vectors vanish at Type I degeneracies ($a = \mathcal{B} = 0$) and at Type II degeneracies ($\varepsilon^\alpha = 0$ because b^α becomes a combination of u^α and ϕ^α). Dividing by a and using $\mathcal{B}/a = \mp\sqrt{\mathcal{E}}$ removes the first problem; writing $\varepsilon^\alpha = g_1 \alpha_1^\alpha + g_2 \alpha_2^\alpha$ in a basis that stays regular, dividing by $\sqrt{g_1^2 + g_2^2}$ and setting $f_{1,2} = 1/\sqrt{2}$ at the degenerate point removes the second (Antón et al., 2010, eqs. 52–67). This is the relativistic version of the procedure of Brio and Wu (1988). The magnetosonic vectors are renormalised similarly (Antón et al., 2010, §3.3.2); their expressions, the left eigenvectors, and the transformation to conserved variables are long and are not reproduced here.

Appendix C

Notation

symbol	meaning
$\rho,\ \epsilon,\ p,\ h$	rest-mass density, specific internal energy, gas pressure, gas specific enthalpy
Γ	adiabatic index of the ideal gas
$\boldsymbol{v},\ W,\ u^\mu$	three-velocity, Lorentz factor, four-velocity
$\boldsymbol{B},\ \boldsymbol{E}$	laboratory magnetic and electric fields ($\sqrt{4\pi}$ absorbed)
$b^\mu,\ b^2$	comoving magnetic field four-vector and its square
η	$\boldsymbol{v}\cdot\boldsymbol{B}$
$w,\ h_t$	total enthalpy density $\rho h + b^2$, total specific enthalpy w/ρ
P	total pressure $p + b^2/2$
σ	magnetisation $b^2/(\rho h)$
$D,\ \boldsymbol{S},\ \tau,\ E$	conserved density, momentum, energy minus rest mass, total energy
$c_s,\ c_a,\ \zeta$	sound speed, rest-frame Alfvén speed $b/\sqrt{w}$, $\sqrt{c_s^2 + c_a^2 - c_s^2 c_a^2}$
$\lambda_\mathrm{e}, \lambda_\mathrm{A}^\pm, \lambda_\mathrm{s}^\pm, \lambda_\mathrm{f}^\pm$	entropy, Alfvén, slow, fast speeds
$a,\ \mathcal{B},\ G$	$u^\mu\phi_\mu$, $b^\mu\phi_\mu$, $\phi^\mu\phi_\mu$ for $\phi_\mu = (-\lambda, 1, 0, 0)$
$V_s,\ W_s,\ n_\mu$	front speed, its Lorentz factor, unit normal $W_s(-V_s, 1, 0, 0)$
$J,\ B_n$	mass flux $-\rho u^\mu n_\mu$, normal comoving field $b^\mu n_\mu$
H	shock invariant $B_n^2/J^2 - b^2/\rho^2$
$\sigma^\mu,\ \boldsymbol{K}$	Alfvén invariant $b^\mu + s\sqrt{w}\,u^\mu$, its spatial ratio $\boldsymbol{\sigma}/\sigma^0$
$[\![Q]\!],\ (\!(Q)\!)$	$Q_a - Q_b$ and $Q_a + Q_b$ (ahead, behind)
ξ	similarity variable x/t
$R_1, \ldots, R_8$	the states of the seven-wave fan
$\boldsymbol{U}^\mathrm{hll},\ \boldsymbol{F}^\mathrm{hll}$	HLL average state and flux

Appendix D

What was verified, and how

The script `sympy_checks.py` in the directory of these notes runs every check below and prints one line per check (about ten seconds on a laptop). Symbolic checks use SymPy; numerical ones use mpmath at 40 significant digits.

Checked symbolically.

- Proposition 2.1: $b^\mu = {}^*F^{\mu\nu}u_\nu$ with $\boldsymbol{E} = -\boldsymbol{v}\times\boldsymbol{B}$ gives (2.5)–(2.7), $u_\mu b^\mu = 0$ and ${}^*F^{\mu\nu} = u^\mu b^\nu - u^\nu b^\mu$.
- The conserved variables and fluxes (2.12)–(2.13) from the components of (2.9), both forms of τ used in `hlld.py`, and the momentum-flux form of Mignone and Bodo (2006).
- The Newtonian limits of Section 2.7.
- The rest-frame dispersion relation: the determinant of the linearised system and of the 2×2 in-plane system; the covariantisation leading to (3.15).
- The identity of (3.15) with the coefficients coded in `rmhd/wave_speeds.py`, with eq. (16) of Del Zanna et al. (2003), and its reductions for $B^x = 0$ and $\boldsymbol{v} = 0$.
- The Alfvén speeds (3.14) and $\mathcal{A} = 0$; the polynomial identities behind Proposition 3.2; the isotropic bound (3.18) and Lemma 3.2.
- The HLL identities; the effective sound speed (7.2); the relativistic Riemann invariant for the ideal gas; the primitive-recovery relations (2.17)–(2.18); the factorisation (5.11).

Checked numerically.

- The eigenvalues and acoustic eigenvectors of one-dimensional relativistic hydrodynamics, (5.4), and with tangential velocity (5.9) and the invariance of $A = hWv_t$.
- Hydrodynamic shocks built from (6.2)–(6.4) satisfy all jump conditions to 10^{-38}; conversely, a shock solved from the jump conditions alone satisfies the Taub adiabat (4.15).
- A fast MHD shock solved from the full RH system satisfies (4.8)–(4.10) and the Lichnerowicz adiabat (4.11) to 10^{-39}, and is an evolutionary $1\to 2$ transition.
- Rotational discontinuities built from Proposition 4.3 satisfy all RH conditions to 10^{-40} at $V_s = \lambda_A$, with ρ, p and b^2 continuous and v_x, $|B_t|$ not.
- The characteristic determinant and the eigenvectors of Chapter B.
- Example 6.1 against the exact solver of the reference implementation (built-in case 3).

Checked against the sources. Every bibliography entry was checked on its arXiv page or through the Crossref record of its DOI (or a publisher or library record for books). Formulas quoted from papers were compared with their text: the quartic and HLLC relations with Mignone and Bodo (2006); the HLLD relations with Mignone et al. (2009); the shock relations, the Lichnerowicz adiabat, the rarefaction ODEs and the test problems with Giacomazzo and Rezzolla (2006); the effective EOS with Romero et al. (2005); the eigenstructure with Antón

et al. (2010); Table 10.1 with Table 1 of Balsara (2001); the rotor with Del Zanna et al. (2003); the Orszag–Tang set-ups with van der Holst et al. (2008) and Beckwith and Stone (2011); the four-quadrant data with Del Zanna and Bucciantini (2002). Statements about the reference implementation were compared with its source and with `HLLD/docs/GUIDE.md` and `HLLD/docs/what_we_solve.md`.

Not verified. (i) The form of the quartic in Leismann et al. (2005), whose full text could not be accessed. (ii) The content of Komissarov (2003) beyond its title, taken from the abstract of the author's preprint on the same subject. (iii) The derivation of the Lichnerowicz adiabat, which was only checked numerically. (iv) The page ranges of Thorne (1973) and Ryu and Jones (1995), for which only the first page is given. (v) Claims attributed to Komissarov (1997) are quoted from the summaries in Mignone and Bodo (2006) and Mignone et al. (2009); the paper itself was not accessed.

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